

Automated Aorta Segmentation and Feature Extraction for Medical Analysis

Ruben Filipe Nascimento Abadesso

Relatório do Projeto de Dissertação
Engenharia Informática
(2º ciclo de estudos)

Orientador: Prof. Dr. Hugo Pedro Martins Carriço Proença

January, 2026

Resumo

As doenças cardiovasculares, particularmente as patologias que afetam a aorta, tais como aneurismas e dissecções, representam uma causa significativa de mortalidade a nível global. A norma clínica atual para diagnóstico e monitorização baseia-se em imagens de tomografia computadorizada, onde a avaliação é frequentemente limitada a medições manuais e unidimensionais, como o diâmetro máximo. Estas métricas podem não capturar totalmente as complexas alterações morfológicas tridimensionais associadas à progressão da doença. Além disso, a segmentação manual da árvore vascular aórtica é um processo de trabalho intensivo e sujeito à variabilidade entre observadores. Esta dissertação apresenta uma estrutura computacional automatizada concebida para segmentar a aorta e extrair características geométricas quantitativas para auxiliar na análise médica.

A metodologia utiliza uma abordagem de aprendizagem profunda para a segmentação volumétrica da aorta e dos seus principais ramos. O sistema é treinado e validado num conjunto de dados multicêntrico composto por 56 exames de angiografia por tomografia computadorizada, caracterizados por parâmetros de aquisição heterogêneos. O pré-processamento de dados envolve a padronização da orientação anatómica, normalização da intensidade e reamostragem para garantir a consistência. A arquitetura de segmentação emprega o *SwinUNETR*, um modelo hierárquico de vision transformer implementado na framework *MONAI*, para capturar características semânticas locais e globais.

Após a segmentação, as máscaras tridimensionais geradas são processadas para criar malhas de superfície. A extração da linha central é realizada utilizando métodos baseados em diagramas de *Voronoi* através do *Vascular Modeling Toolkit*. Esta abstração geométrica serve como referência para o cálculo de descritores morfológicos, incluindo o diâmetro, curvatura, torção e tortuosidade ao longo do eixo longitudinal. Os resultados experimentais indicam que o modelo de segmentação atinge um coeficiente de similaridade de *Dice* médio de 0,93 no conjunto de teste. A estrutura desenvolvida fornece um fluxo de trabalho reprodutível para a extração de descritores de forma avançados, oferecendo dados quantitativos que podem complementar as métricas clínicas tradicionais na avaliação da morfologia aórtica.

Palavras-chave

Segmentação da Aorta, Aprendizagem Profunda, SwinUNETR, Extração de Características Geométricas, Angiografia por Tomografia Computorizada, Análise de Imagem Médica

Resumo alargado

As doenças cardiovasculares constituem atualmente a principal causa de mortalidade a nível global, representando um desafio significativo para os sistemas de saúde e para a investigação médica. Entre as diversas patologias vasculares, as condições que afetam a aorta — a principal artéria sistémica responsável pela distribuição de sangue oxigenado — assumem particular relevância devido à sua natureza potencialmente fatal. A integridade estrutural e morfológica da aorta é fundamental para a estabilidade hemodinâmica, contudo, este vaso é propenso a remodelações patológicas graves, nomeadamente aneurismas da aorta e dissecções da aorta.

Na prática clínica atual, o diagnóstico e a monitorização destas patologias baseiam-se essencialmente na análise de imagens de angiografia por tomografia computadorizada. O processo de decisão para intervenção cirúrgica depende frequentemente da medição manual de biomarcadores específicos, com destaque para o diâmetro máximo. Embora o diâmetro seja uma métrica fundamental, a dependência de medições unidimensionais pode não capturar adequadamente as alterações morfológicas complexas e tridimensionais associadas à progressão da doença, tais como a dilatação assimétrica ou a tortuosidade irregular do vaso. Adicionalmente, o processo de segmentação e análise manual de exames de tomografia computadorizada é moroso e trabalhoso, estando sujeito à variabilidade inter-observador, o que pode conduzir a inconsistências no diagnóstico e no acompanhamento longitudinal dos pacientes.

Neste contexto, o presente trabalho de dissertação propõe o desenvolvimento de um sistema computacional automatizado destinado à segmentação da aorta e à extração de características geométricas para análise médica. O objetivo primordial é colmatar a lacuna existente entre os dados radiológicos brutos e a informação clínica acionável, fornecendo ferramentas que permitam uma caracterização quantitativa mais abrangente da geometria da aorta. O sistema proposto visa não apenas a segmentação precisa da aorta e dos seus principais ramos — incluindo o tronco braquiocéfálico, as artérias carótidas comuns, as artérias subclávias e as artérias ilíacas comuns — mas também a geração de malhas de superfície de alta fidelidade para a análise numérica da morfologia do vaso.

A metodologia adotada utiliza o conjunto de dados *"Aortic Vessel Tree"*, disponibilizado no âmbito do desafio *MICCAI SEG.A. 2023*. Este conjunto de dados caracteriza-se pela sua heterogeneidade, incluindo 56 exames de angiografia por tomografia computadorizada provenientes de três instituições distintas, apresentando variações significativas em termos de protocolos de aquisição, resolução espacial e níveis de ruído. O processo computacional inicia-se com uma fase de pré-processamento rigoroso, que inclui a reorientação dos dados para uma posição anatómica padrão, a normalização dos valores de intensidade (unidades *Hounsfield*) e a reamostragem para uma dimensão de voxel padronizada de $1 \times 1 \times 1 \text{ mm}$, garantindo a consistência espacial necessária para os modelos de aprendizagem profunda subsequentes.

Para a tarefa de segmentação semântica, o estudo recorre à biblioteca *MONAI*, selecionando a arquitetura *SwinUNETR*. Esta arquitetura híbrida combina a estrutura hierárquica de um *Vision Transformer* com um decodificador em forma de "U" (*U-Net*). A escolha desta abordagem justifica-se pela capacidade dos *Transformers* em modelar dependências de longo alcance e capturar o contexto global da anatomia volumétrica, superando as limitações de campo recetivo das redes neurais convolucionais tradicionais. Durante a fase de treino e validação, o modelo demonstrou um desempenho robusto, alcançando um coeficiente de similaridade *Dice* de 0,917 na validação e uma média de 0,93 no conjunto de teste, evidenciando a sua eficácia na delineação da árvore vascular aórtica.

Após a obtenção das máscaras volumétricas, o sistema procede à análise geométrica utilizando o *Vascular Modeling Toolkit*. As máscaras binárias são convertidas em malhas de superfície, que servem de base para a extração da linha central do vaso através de algoritmos baseados em diagramas de *Voronoi*. A partir desta representação esquelética e da superfície do vaso, são calculados diversos descritores morfológicos quantitativos ao longo do eixo longitudinal da aorta. Estes parâmetros incluem o diâmetro, a curvatura local, a torção e a tortuosidade. A análise permite a criação de gráficos detalhados que ilustram a variação destas métricas ao longo da linha central e nas respetivas ramificações, possibilitando a deteção de anomalias geométricas, que métricas globais simples poderiam omitir.

Os resultados preliminares indicam que a abordagem proposta é capaz de automatizar com sucesso todo o fluxo de trabalho, desde a entrada da imagem médica até à geração de relatórios morfológicos. A integração de arquiteturas de estado da arte, como o *SwinUNETR*, com algoritmos robustos de modelação geométrica, permite uma avaliação mais detalhada da condição aórtica. O trabalho futuro, delineado no planeamento da dissertação, prevê a continuação do desenvolvimento do sistema, a comparação com outros modelos de referência e o refinamento dos algoritmos de extração de características, visando a consolidação de uma ferramenta de apoio à decisão clínica que facilite a avaliação de risco e o planeamento cirúrgico em patologias aórticas.

Abstract

Cardiovascular diseases, particularly pathologies affecting the aorta such as aneurysms and dissections, represent a significant cause of mortality globally. The current clinical standard for diagnosis and monitoring relies on Computed Tomography (CT) imaging, where assessment is often limited to manual, unidimensional measurements such as the maximum diameter. These metrics may not fully capture the complex three-dimensional morphological changes associated with disease progression. Furthermore, manual segmentation of the aortic vessel tree is a labor-intensive process subject to inter-observer variability. This dissertation presents an automated computational framework designed to segment the aorta and extracting quantitative geometric features to assist in medical analysis.

The methodology utilizes a deep learning approach for the volumetric segmentation of the aorta and its principal branches. The system is trained and validated on a multicenter dataset comprising 56 Computed Tomography Angiography (CTA) scans, characterized by heterogeneous acquisition parameters. Data preprocessing involves the standardization of anatomical orientation, intensity normalization, and resampling to ensure consistency. The segmentation architecture employs *SwinUNETR*, a hierarchical vision transformer model implemented within the Medical Open Network for AI (MONAI) framework, to capture both local and global semantic features.

Following segmentation, the generated three-dimensional masks are processed to create surface meshes. Centerline extraction is performed using *Voronoi* diagram-based methods via the Vascular Modeling Toolkit (VMTK). This geometric abstraction serves as a reference for computing morphological descriptors, including vessel diameter, curvature, torsion, and tortuosity along the longitudinal axis. Experimental results indicate that the segmentation model achieves an average Dice Similarity Coefficient (DSC) of 0.93 on the test set. The developed framework provides a reproducible pipeline for the extraction of advanced shape descriptors, offering quantitative data that may complement traditional clinical metrics in the evaluation of aortic morphology.

Keywords

Aorta Segmentation, Deep Learning, SwinUNETR, Geometric Feature Extraction, Computed Tomography Angiography, Medical Image Analysis

Contents

1	Introduction	1
1.1	Context and Motivation	1
1.2	Objectives	2
1.3	Problem Statement	2
1.4	Dissertation Structure	3
2	Related Work	5
2.1	Medical Context	5
2.1.1	Anatomy of the Aorta	5
2.1.2	Pathologies	6
2.1.3	Clinical Relevance of Geometric Features	7
2.1.4	Imaging Modalities	10
2.2	Deep Learning Fundamentals	11
2.2.1	Convolutional Neural Networks	11
2.2.2	Encoder-Decoder	13
2.2.3	U-Shaped Convolutional Networks	14
2.2.4	Vision Transformers	16
2.2.5	Hybrid and Transformer-based Architectures	18
2.3	State of the Art in Medical Segmentation	20
2.3.1	Evolution of Computational Methods for Aortic Analysis	20
2.3.2	MICCAI SEG.A. 2023 Challenge	22
2.3.3	MONAI Framework	23
2.4	Geometric Analysis and Feature Extraction	24
2.4.1	Centerline Extraction and Skeletonization	24
2.4.2	Cross-Sectional and Diameter Assessment	25
2.4.3	Surface Mesh Generation and Volumetric Analysis	26
2.4.4	Tortuosity, Curvature, and Advanced Shape Descriptors	26
2.5	Summary and Discussion	27
3	Proposed Method	29
3.1	Dataset Description	29
3.2	System Overview	30
4	Work Planification	35
	Bibliography	37

List of Figures

2.1	CNN architecture for image classification.	12
2.2	3D CNN architecture for image classification.	12
2.3	Encoder-Decoder architecture.	14
2.4	U-Net architecture.	15
2.5	3D U-Net architecture.	16
2.6	Vision Transformer architecture.	17
2.7	UNETR architecture.	19
2.8	SwinUNETR architecture.	20
3.1	Distribution of slice thickness for each CT scan. Scans are colored by site. . . .	29
3.2	Pixel level histogram. Scans are colored by site. This shows that scans from different sites do not have similar distribution.	30
3.3	Example of the AVT data.	30
3.4	Diagram overview of the method’s framework.	31
3.5	Validation Performance (Dice Score) during <i>SwinUNETR</i> segmentation train- ing.	31
3.6	Aorta diameter measurement along the centerline of the vessel, and respective branches.	32
3.7	Aorta curvature measurement along the centerline of the vessel, and respec- tive branches.	32
3.8	Aorta torsion measurement along the centerline of the vessel, and respective branches.	33
3.9	Aorta tortuosity measurement of each segment of the vessel. A new segment is created at every bifurcation. The respective tortuosity value of each segment is calculated based on the respective start and end point.	33
4.1	Gantt chart illustrating the development timeline and duration of the pro- posed tasks.	35

List of Tables

- 2.1 Image information of the aortic vessel trees. Number of axial slices, segmentation times, slice thickness and vessel tree volume are given as: min/median/max. 23

Acronyms

AA	Aortic Aneurysm
AAA	Abdominal Aortic Aneurysm
AD	Aortic Dissection
AVT	Aortic Vessel Tree
CNN	Convolutional Neural Network
CT	Computed Tomography
CTA	Computed Tomography Angiography
DSC	Dice Similarity Coefficient
DICOM	Digital Imaging and Communications in Medicine
ESVS	European Society for Vascular Surgery
EVAR	Endovascular Aneurysm Repair
GELU	Gaussian Error Linear Unit
HD95	95th percentile of the Hausdorff Distance
HU	Hounsfield Unit
ILT	Intraluminal Thrombus
IVUS	Intravascular Ultrasound
MICCAI	Medical Image Computing and Computer Assisted Intervention
MONAI	Medical Open Network for AI
MRI	Magnetic Resonance Imaging
NIFTI	Neuroimaging Informatics Technology Initiative
ReLU	Rectified Linear Unit
SEG.A.	Segmentation of the Aorta
TAA	Thoracic Aortic Aneurysm
VMTK	Vascular Modeling Toolkit
ViT	Vision Transformer
2D	Two Dimensional
3D	Three Dimensional

Chapter 1

Introduction

1.1 Context and Motivation

Cardiovascular diseases remain the main cause of mortality globally, presenting a serious challenge for healthcare systems and medical research. Of all the conditions, pathologies affecting the aorta, which is the main systemic artery responsible for delivering oxygenated blood to the entire body, are of particular significance because of their potentially life-threatening nature. The structural integrity and morphological state of the aorta is vital in ensuring hemodynamic stability, but this artery is particularly prone to serious pathological remodeling, most notably Aortic Aneurysms (AAs) and Aortic Dissections (ADs).

Currently, the diagnosis of aortic pathologies essentially involves the use of CT scans. In clinical practice, the decision-making process for intervention has often involved the manual measurement of particular biomarkers, specifically the maximum diameter of the vessel. While diameter is a fundamental metric, relying on unidimensional measurements may not fully capture the complex, Three Dimensional (3D) morphological changes associated with disease progression. For instance, asymmetric dilation or irregular vessel tortuosity can significantly influence hemodynamic stress, yet these factors are difficult to quantify precisely through manual inspection alone.

In addition, the process of manual segmentation and analysis of CT scans is a labor-intensive and time-consuming process. Radiologists are faced with the challenge of dealing with large volumetric datasets to establish the boundaries of the blood vessels in order to isolate the aorta from surrounding anatomical structures. This manual approach is also prone to inter-observer variability, potentially leading to inconsistencies in diagnosis and follow-up monitoring.

With this challenges in mind, there is an increasing need for automated computational frameworks that can efficiently process images with high accuracy. By leveraging advances in deep learning and computer vision, it is possible to develop systems that automatically segment the aorta and extract a large number of geometric descriptors. These automated morphological measurements are not intended to replace standard clinical metrics but rather to complement them, providing clinicians with a more comprehensive quantitative characterization of the vessel's geometry. Such tools have the potential to enhance the understanding of aortic remodeling and support more informed medical decisions.

1.2 Objectives

The primary objective of this dissertation is to address the current limited availability of comprehensive computational tools for the analysis of cardiovascular pathologies, specifically AA and AD. The research focuses on the development of a fully automated system designed to assist in clinical evaluation and the longitudinal monitoring of disease progression. The core methodology entails the precise segmentation of the aorta and its principal branches, including the brachiocephalic trunk, common carotid arteries, subclavian arteries and common iliac arteries (encompassing both their internal and external segments). Leveraging the resulting 3D segmentation masks, the system generates high-fidelity surface meshes to extract critical geometric features. This process aims to provide clinicians with both advanced visual representations and rigorous numerical analyses of the vessel’s morphology, ultimately delivering an automated software solution that streamlines routine follow-up examinations and facilitates the tracking of pathological remodeling.

To successfully achieve these goals, a structured methodological pipeline is established. Initially, patient CT scans undergo rigorous preprocessing to standardize Hounsfield Unit (HU), anatomical orientation, and voxel spacing. This foundational step ensures data consistency and optimal radiodensity distribution for the subsequent deep learning models. Following this, state-of-the-art segmentation architectures are evaluated using the prepared dataset to identify the most effective computational method. Upon selection, the generated 3D volumetric masks are converted into surface meshes, which serve as the basis for extracting quantitative geometric descriptors.

By addressing these objectives, this research aims to advance the field of computer-aided diagnosis by bridging the gap between raw radiological data and actionable clinical insights, thereby enhancing the precision of aortic risk assessment and supporting more informed therapeutic decision-making.

1.3 Problem Statement

However, the development of advanced computerized frameworks for aortic analysis is hindered by data availability and consistency. Due to strict rules regarding patient data privacy, large amounts of publicly accessible data with annotated computed tomography scans are not easily obtained. This limits the amount of training data that can be used for deep learning methods, especially when compared to tasks in the domain of computer vision in general. Furthermore, a high level of diversity can be found among existing data sets. CT scans are found to vary in resolution, slice thickness, as well as noise levels due to differences in acquisition protocols as well as manufacturers.

1.4 Dissertation Structure

The present document is structured as follows:

- **Chapter 1 - Introduction:** Presents the research context and objectives, and identifies the specific challenges encountered.
- **Chapter 2 - Related Work:** Provides the medical and theoretical background relevant to the study, including an overview of aortic anatomy and computational methods for medical analysis. It also reviews state-of-the-art techniques for aortic segmentation and geometric feature extraction from surface meshes.
- **Chapter 3 - Proposed Method:** Details the proposed framework for the segmentation of the aorta, the extraction and analysis of geometric features, and the evaluation of the vessel's condition.
- **Chapter 4 - Work Planification:** Outlines the project schedule and the timeline associated with the completion of the defined tasks.

Chapter 2

Related Work

2.1 Medical Context

The aorta is the major systemic artery in the human body, from which the entire arterial system arises, and plays the crucial role of maintaining the physiological functionality of the circulatory system. Understanding its anatomy and the pathologies that change its morphology is the key to developing automated segmentation and feature extraction systems. This section provides an overview on aortic anatomy, the cardiovascular diseases that require medical imaging intervention and the morphological biomarkers clinicians use for diagnosis and surgical planing.

2.1.1 Anatomy of the Aorta

The aorta serves as the main vessel of the systemic arterial circulation, originating from the left ventricle of the heart. It is responsible for distributing oxygenated blood to the tissues of the body. The aorta is divided into four distinct segments, each with specific geometric characteristics and susceptibility to different diseases:

Ascending Aorta

This segment starts in the base of the left ventricle. It ascends superiorly, slightly forwards and to the right, within the pericardial cavity. The initial portion marks the aortic root where the aortic valve and coronary arteries originate. This part of the aorta is very susceptible to dilatation especially when it comes to connective tissue diseases [1].

Aortic Arch

After the ascending aorta, the aortic arch curves backward and left, forming an arch over the bifurcation of the pulmonary artery and the left main bronchus. This portion of the aorta is geometrically complex because of its high curvature and the presence of supra-aortic arteries, like the brachiocephalic trunk, left common carotid artery and left subclavian artery. The geometric complexity of the aortic arch results in particular hemodynamic shear forces, which contribute to the development of atherosclerosis [2].

Descending Thoracic Aorta

Continuing from the end of the aortic arch until the level of the diaphragm, passing through the posterior mediastinum, this segment is comparatively straight compared with the arch, although it often becomes very tortuous with age and hypertension [3].

Abdominal Aorta

After crossing the diaphragm the aorta enters the abdominal cavity. It continues in front of the lumbar vertebrae, ending by bifurcating into the left and right common iliac arteries, at the level of the fourth lumbar vertebra. This portion of the aorta is the most common site for aneurysms.

The aortic wall is composed of three layers: the *intima* (inner layer), the *media* (muscular middle layer), and the *adventitia* (outer connective layer) [4]. The mechanical integrity of the aorta relies heavily on the elastin and collagen fibers within the *media*. Degeneration or failure of these layers leads to the morphological abnormalities, such as dilation or delamination, that automated systems aim to detect through CT imaging.

2.1.2 Pathologies

Cardiovascular diseases involving the aorta can be dangerous and generally occur because of alterations in the geometry of the vessel. The primary focus of computational analysis in this research centers on AAs and ADs.

Abdominal Aortic Aneurysm

An aneurysm is defined as a localized dilatation of an artery having at least a 50% increase in diameter compared to standard size. In the case of the abdominal aorta, a diameter above 3.0 cm is clinically recognized as a aneurysm [5]. Because of the absence of symptoms, Abdominal Aortic Aneurysms (AAAs) have come to be recognized as "silent killers", as they can grow progressively until rupture occurs. Ruptured AAAs have a mortality rate of over 80% [5], making early detection and surveillance critical.

Pathophysiologically, AAA is the consequence of the degradation of the extracellular matrix and the loss of elastin [6], resulting in the weakening and dilation of the aorta due to intraluminal pressure. This dilation rarely occurs symmetrically. AAAs often present with complex, non-circular cross-sections and varying degrees of asymmetry. Clinicians currently rely on the maximum diameter as the primary criterion for surgical intervention (typically >5.0 cm for women and >5.5 cm for men) [5]. Though it is well understood that diameter alone is an insufficient predictor of rupture risk and more complex geometry, like volume and the curvature of the surface, needs to be evaluated.

Thoracic Aortic Aneurysm

While less common than AAAs, Thoracic Aortic Aneurysms (TAAs) have significant segmentation challenges due to the proximity to the heart and the complex branches of the aortic arch. TAAs are often linked with conditions like Marfan syndrome, Ehlers-Danlos syndrome or atherosclerosis [7]. The geometrical evaluation of TAAs is very important since the complex blood flow patterns within the thoracic cage may contribute to high rates of expansion.

Aortic Dissection

AD is a life-threatening medical emergency caused by a tear in the *intima* layer of the aortic wall. This tear allows blood to flow into the *media*, forcing the layers apart. This separation creates an internal membrane, referred to as intimal or dissection flap, that divides the aorta into two separate pathways of blood flow: the true lumen (the original blood channel) and the false lumen (the newly formed channel within the wall). A serious medical complication is possible if blood is separated from flowing inside the aorta. A tear in the aorta can be a consequence of a wide range of factors, such as chronic high blood pressure, connective tissue disorders, atherosclerosis, aging and smoking.

From both geometric and segmentation standpoints, ADs are very complex. The separating flap (the intimal flap) can twist spirally along the aorta, creating complex topologies that vary slice by slice in a CT scan. The Stanford classification system divides dissections into Type A (involving the ascending aorta) and Type B (descending aorta) [8]. Type A requires immediate surgery, while Type B is usually treated either by medication or via endovascular repair in more complicated cases. Automated systems must be capable of distinguishing the exterior boundary of the aorta from the internal structures to accurately calculate the total vessel volume and the compression of the true lumen.

Atherosclerosis and Calcification

Atherosclerosis is the buildup of plaque (fats, cholesterol and calcium) in the arterial walls. In CT scans, the calcified plaque show up as high-intensity regions. Although not considered a geometric abnormality of the external boundary of the aorta, calcification does affect the compliance of the vessel and is a major risk factor for the development of aneurysms and dissections. A high concentration of calcification may affect the curvature of the aorta and its eccentricity, thereby disrupting laminar flow and inducing shearing forces against the walls of the vessels.

2.1.3 Clinical Relevance of Geometric Features

Although the segmentation of the aorta gives a good graphic visualization of its anatomy, additional quantitative analysis of the morphological measurements may provide complementary information beyond standard clinical metrics. Clinical decision-making has traditionally been dependent on unidimensional measurements, like maximum diameter, despite the aorta's inherently 3D geometry and its susceptibility to pathological remodeling. In this context, a range of geometric features can be computed from segmented volumes to describe shape and spatial organization more comprehensively. Although these features are not yet part of routine clinical practice, they have been explored as potential imaging biomarkers for improving risk stratification in aortic diseases.

Maximum Diameter

The maximum aortic diameter is still the "gold standard" marker that defines the need for

surgical intervention. According to guidelines from the European Society for Vascular Surgery (ESVS), elective repair for AAAs is recommended when the diameter exceeds 5.5 cm in men and 5.0 cm in women, or if the expansion rate exceeds 1 cm per year [5].

Despite its widespread use, the maximum diameter has limitations. The maximum diameter calculation assumes that an aneurysm expands evenly and does not take into consideration any possible localized bulges or "blebs", that may be susceptible to rupture despite a smaller overall diameter. In addition, manually measuring aneurysm's diameter on a Two Dimensional (2D) axial image can lead to errors if the aorta is tortuous and not perpendicular to the scan plane. Automated centerline-based analysis allows for the calculation of the maximum diameter orthogonal to the centerline, thereby providing a much more accurate estimate of aneurysm dimensions.

Aortic Volume

Volumetric analysis provides a global quantification of the aneurysm that diameter measurements often miss. Research suggests that aneurysms can grow in volume without substantial change in maximum diameter [9]. Total vessel volume includes both the blood-filled lumen and the volume of the Intraluminal Thrombus (ILT), a fibrin structure that sits along the inside of the aneurysm wall.

The ratio between the volume of the lumen and the total volume is an important hemodynamic parameter. The measurement of the rate of volumetric growth is increasingly viewed as a more sensitive predictor of rupture risk assessment than measurement of diameter alone [10, 11]. Automated segmentation allows for the precise integration of voxel volumes, removing the inter-observer variability inherent in manual geometric approximations.

Tortuosity

Tortuosity indicates the degree of twisting or turning of the vessel path. It is clinically significant for hemodynamics and surgical planning.

When the aorta is highly tortuous, it causes disturbed flow patterns, which may include flow separation and retrograde vortices. These flow disturbances alter wall shear stress, a frictional force that affects endothelial cell function [2]. Areas of low or oscillating wall shear stress are linked with the development of atherosclerotic plaque [12].

In Endovascular Aneurysm Repair (EVAR), a stent-graft is inserted through the femoral arteries to bypass the aneurysm. However, significant tortuosity within the iliac arteries or the aortic neck (the region just below the renal arteries) can make the navigation and deployment of these rigid devices difficult or impossible.

Tortuosity is mathematically quantified using the Distance Metric (ratio of the centerline length to the Euclidean distance between endpoints) or curvature-based metrics.

Curvature and Torsion

While tortuosity describes the vessel globally, curvature and torsion describe the local geometry of the centerline.

Curvature measures the degree to which the vessel deviates from a straight line at a specific point. High curvature in the aortic arch is a natural physiological feature, but excessive curvature in the abdominal aorta may indicate abnormal tortuosity and be associated with adverse hemodynamic conditions. High curvature regions experience higher mechanical stress on the outer side, making the intima there more susceptible to tears that can lead to dissections.

Torsion, describes how much the aortic centerline twists along its length, reflecting the 3D geometry of the vessel. Higher torsion values indicate a more helical or spiral shape. This is particularly relevant in Type B ADs, where the twisting geometry of the aorta can influence the course and shape of the dissection flap, although other factors such as blood flow and wall stress also play a role.

Eccentricity and Ellipticity

Assuming the aorta is a perfect cylinder is a mathematical simplification that rarely reflects pathological anatomy. Eccentricity, quantifies how much a cross-section deviates from a circle. Ellipticity specifically models the cross-section as an ellipse, defined by the ratio of the major axis to the minor axis.

In AAAs, the vessel often remodels asymmetrically. According to the Law of Laplace, wall tension is proportional to the radius [13]. However, in non-circular vessels, stress concentrates in areas of high local curvature (small radius). Therefore, an aneurysm with a highly elliptical or irregular cross-section may experience peak wall stresses significantly higher than a circular aneurysm of the same equivalent diameter. Quantifying eccentricity and ellipticity provides a more accurate estimation of local wall stress, which can improve rupture risk assessment beyond diameter-based criteria.

Surface Area

The surface area of the aorta can be used as a global geometric descriptor that reflects the enlargement and remodeling of the vessel. In cases such as Marfan syndrome, dilation of the aortic root leads to an increase in surface area as a consequence of increasing radius, accompanying the rise in wall stress driven primarily by diameter, pressure, and wall thickness. In addition to size, the surface area-to-volume ratio provides a quantitative descriptor of aneurysm shape: compact geometries such as saccular aneurysms approach spherical configurations with lower surface area for a given volume, whereas fusiform, elongated aneurysms exhibit higher ratios. Distinguishing between saccular and fusiform morphologies is clinically important, as saccular aneurysms are often considered higher risk and are treated at

smaller sizes, likely due to localized wall weakness and asymmetric stress distributions rather than global size metrics alone.

2.1.4 Imaging Modalities

Medical imaging plays a vital role in the diagnosis, monitoring, and treatment of cardiovascular diseases. Although various imaging techniques, including Magnetic Resonance Imaging (MRI) and Ultrasound, have been employed in clinical practice, CT imaging is the most commonly used imaging modality to assess aortic diseases owing to its superior resolution capabilities. The processing of medical imaging data requires standardized image formats to enable interoperability between clinical imaging devices and computational processing tools.

Computed Tomography and Angiography

CT imaging measures the attenuation of X-rays as they pass through human body tissues. The attenuation of X-rays varies according to tissue composition. High-density tissues, including bones or calcified tissues, have higher attenuation coefficients compared to low-density tissues such as air or fatty tissues. In a CT image, high-density tissues appear brighter compared to low-density tissues. A CT imaging system consists of a detector that measures various projections of X-rays transmitted from various angles. The detector then uses mathematical algorithms to reconstruct a 3D image of the body. The reconstructed image consists of a stack of axial slices.

However, non-contrast CT imaging may not always provide satisfactory results for vascular imaging owing to the low contrast between the blood pool and the surrounding soft tissues. CTA imaging provides better visualization of the vasculature. In CTA imaging, a contrast agent is injected intravenously to increase the attenuation coefficients of the blood pool. The increase in attenuation coefficients of the blood pool provides better visualization of the vessel lumen. The vessel lumen is clearly differentiated from the vessel wall. In the case of aortic aneurysm and dissection, CTA imaging plays a vital role in imaging the vessel curvature, identifying ILT, and detecting dissection flaps between true and false lumens

DICOM and NIfTI

For the management and processing of medical images, specific data formats are used. In a hospital setup, the standard format for storing and transferring medical imaging data is the Digital Imaging and Communications in Medicine (DICOM) format. The format combines the pixel values of the medical image with a significant amount of data, including patient information, scanning parameters, and machine settings. Although the format guarantees excellent interoperability between hardware and software in a hospital setup, it is not appropriate for deep learning processing, since the data is generally stored as a series of individual 2D files, one for each axial slice of a volumetric scan.

In order to facilitate the processing of medical image data for deep learning, the data is gener-

ally converted from the DICOM format to the Neuroimaging Informatics Technology Initiative (NIfTI) format. The NIfTI format is generally accepted by the medical image computing community, since it allows for the consolidation of the entire 3D data set into a single file. The format includes a lightweight header with minimal information, such as spatial information, which is essential for converting voxel coordinates to physical coordinates. In addition, the conversion from the DICOM format to the NIfTI format is generally accompanied by the removal of personal health information, ensuring data anonymization.

2.2 Deep Learning Fundamentals

The field of automated analysis in medical imaging has undergone a paradigm shift with the development of deep learning, a branch of machine learning that uses multi-layered neural networks. These methods also have the potential for the automated discovery of sophisticated feature representations from raw input data in the form of pixel or voxel intensities in CT images. In the area of cardiovascular imaging analysis, deep learning methods are used for anatomical structure segmentation and the identification of pathologies using the resulted features. This section introduces the fundamental architectures that underlie modern segmentation techniques: Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) [14].

2.2.1 Convolutional Neural Networks

CNNs are the building blocks of many computer vision applications such as image classification, object detection, and semantic segmentation. Inspired by the architecture of the visual cortex, CNNs are designed to process data with a grid structure. For instance, unlike fully connected networks where every neuron connects to every neuron in the next layer, CNNs selectively use local and shared weights. This architectural choice makes the network translation-equivariant, meaning that it can pick out a given feature (edges or textures) regardless of where it appears spatially in the input.

The basic component of a CNN is the convolutional layer. A convolutional layer works by moving a set of learnable filters or kernels over the input image. At each point, an element-wise multiplication operation is done between the filter values and the input values within the receptive field and then summing the results to get a single value for the output feature map. In the earlier layers of the network, the filters will normally learn to extract information pertaining to basic geometric information including edges, corners, and curves. As the information is propagated deeper into the network, the network attempts to extract more complex information such as the aortic wall or lumen by combining the basic geometric information.

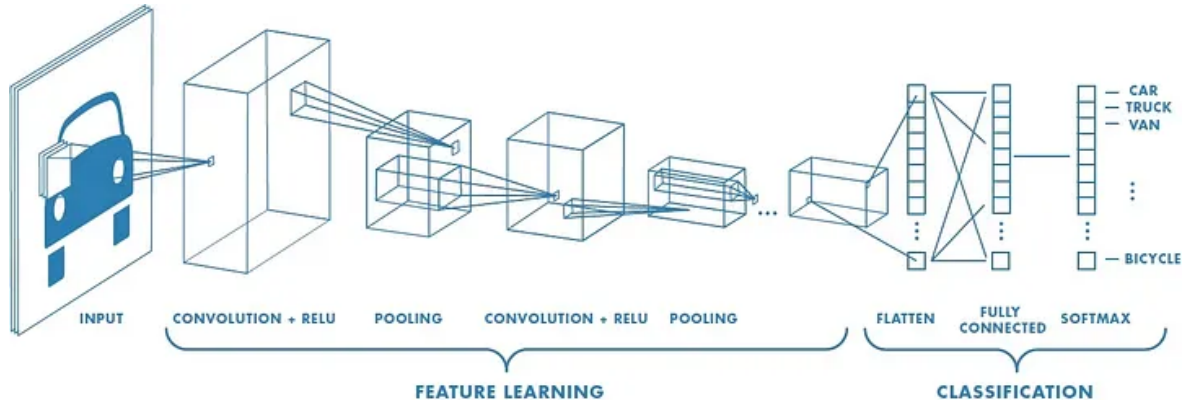


Figure 2.1: CNN architecture for image classification.

After the convolutional layers, activation functions are introduced to the feature maps. Functions such as the Rectified Linear Unit (ReLU) activation function or its variants, such as the Gaussian Error Linear Unit (GELU) activation function, are used to add non-linearity to the model. Additionally, pooling layers are often used between convolutional layers to successively lower the spatial resolution of the feature map layers. Operations such as max pooling or average pooling are used with the primary goals of increasing efficiency and also to add a sense of translation invariance to feature maps, thereby increasing resistance to minute changes in position of anatomical structures.

3D Convolutions in Medical Imaging

Though conventional CNNs handle 2D images, medical images, specifically CTs angiographies, are inherently 3D volumes. Handling these volumes slice by slice, which involves 2D analysis, can lead to a lack of spatial continuity in the z-axis (longitudinal direction), which can cause inconsistency in segmentation between slices. To handle this, 3D convolutions are used. In 3D CNNs, the filters move through three dimensions: height, width, and depth. They learn features by capturing volumetric information.

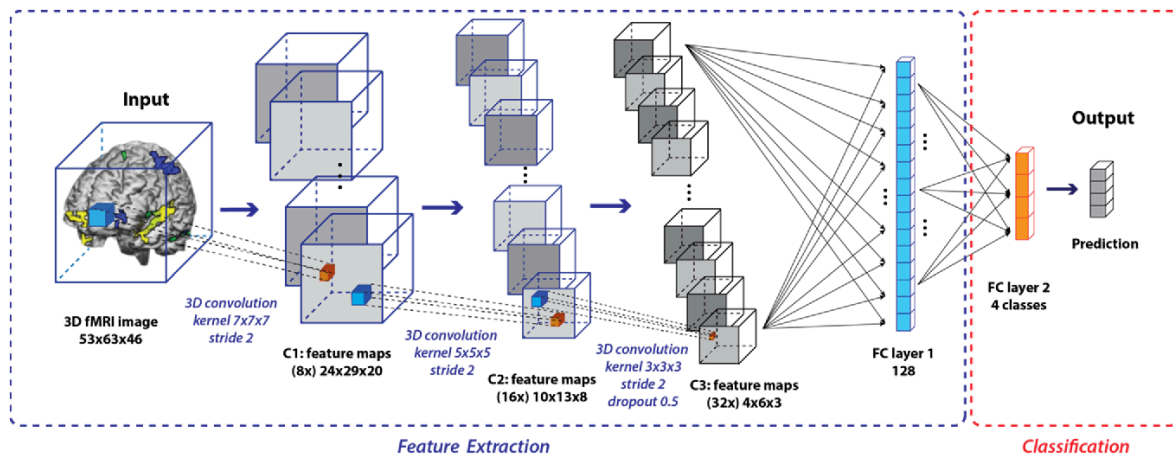


Figure 2.2: 3D CNN architecture for image classification.

The use of 3D convolution algorithms is specifically relevant in the context of aorta segmen-

tation. This is because the aorta itself is a homogeneous, tubular structure which extends from the thoracic region into the abdominal region in a tortuous fashion in most cases. In this case, the 3D convolution algorithm becomes useful in the sense that it enables the network to utilize information between slices in order to accurately separate the structure from the surrounding tissues of similar radiodensity, such as the *vena cava* or muscle tissue. However, the use of 3D CNN algorithms becomes associated with higher computational costs compared to 2D CNN algorithms, necessitating efficient architectural designs and often requiring the use of patch-based training strategies where the network processes smaller volumetric crops rather than the entire scan simultaneously.

Despite their effectiveness, CNNs have limitations regarding the receptive field. This is due to the localized nature of the convolution process, which means that an individual neuron is only able to process the input from a small region. Although the process of stacking multiple layers helps to improve the receptive field, the ability to process long-distance dependencies effectively is still an impediment for CNNs. This is an important consideration in the case of the aorta, where the connection between the aortic arch and the iliac bifurcation region offers vital context.

2.2.2 Encoder-Decoder

The original application of CNNs was in image classification tasks, where the task is to classify a whole image according to a single label. However, in medical image analysis tasks, a higher level of understanding is also needed. This is because, in the case of extraction of geometric information such as aortic diameter, volume, or tortuosity, semantic segmentation is needed. This is a task where the network has to classify each pixel or voxel in the image according to a label. This helps separate the region of interest from the background. This task is made possible by the use of the encoder-decoder architecture.

An encoder-decoder network consists of two separate pathways that perform complementary functions. The first pathway is the encoder or contracting pathway, and functions like a traditional classification network. It is made up of multiple layers of convolutional layers followed by downsampling layers such as max-pooling or strided convolutions. The main goal of the encoder is to gradually decrease the spatial dimensions of the input feature maps while also increasing the depth of the feature maps. As the information is passed through the pathway, it is forced to ignore the high spatial frequencies while attempting to learn a semantic representation that defines "what" is in the image rather than "where" it is.

The result of the encoder is the low-resolution, high-dimensional feature representation, which is usually denoted as the bottleneck or latent space. While the compressed features in the bottleneck retain the rich semantic information, they lack the spatial resolution to describe the boundaries of the anatomy like the aorta. This issue will be resolved by the addition of the decoder, also called the expanding path. The purpose of the decoder is to enhance the

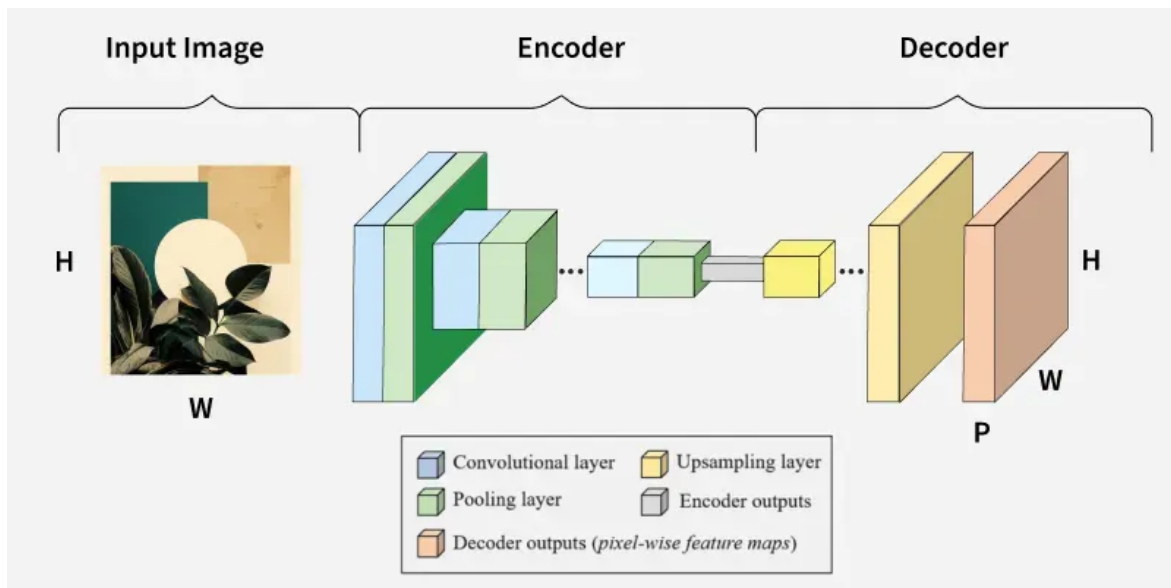


Figure 2.3: Encoder-Decoder architecture.

spatial resolution of the data, making it similar in resolution to that of the input, by utilizing transposed convolution or bilinear upsampling layers, followed by convolutional layers.

So, the decoder's job is to bring back the spatial detail that got lost when the encoder did its downsampling. While progressively upsampling the feature maps, it carries the semantic essence stored in the bottleneck and aligns it with the exact pixel or voxel locations. In the last layer, you normally see a 1×1 convolution (or $1 \times 1 \times 1$ in 3D), followed by a softmax or sigmoid, which gives you a probability map, telling how likely each element is part of the foreground, effectively creating the segmentation mask. The overall design maintains a balance between contextual understanding and precise localization, important for medical tasks, where both the organ's identity and its actual shape are relevant for further geometric analysis.

2.2.3 U-Shaped Convolutional Networks

While the generic encoder-decoder setup marked a big advance in dense prediction, early versions often produced segmentation masks with blurry boundaries. This is because the downsampling of the encoder alone has lost fine-grained spatial details which upsampling alone in the decoder cannot fully recover. Given that accurately tracing vessel walls in medical imaging is quite necessary to measure features such as eccentricity or surface curvature, edge precision is a must. To address this loss of information, *Ronneberger et al.* [15] proposed the *U-Net*, which has since become the standard for image segmentation.

U-Net

The name *U-Net* comes from its symmetric U-shaped structure, which consists of a contracting encoder on the left side and an expanding decoder on the right. The main novelty is represented by the skip connections that link both sides. These connections take feature maps at

high-resolution from a given encoder layer and merge them directly with the corresponding upsampled maps in the decoder, preserving fine detail while the network reconstructs the segmentation.

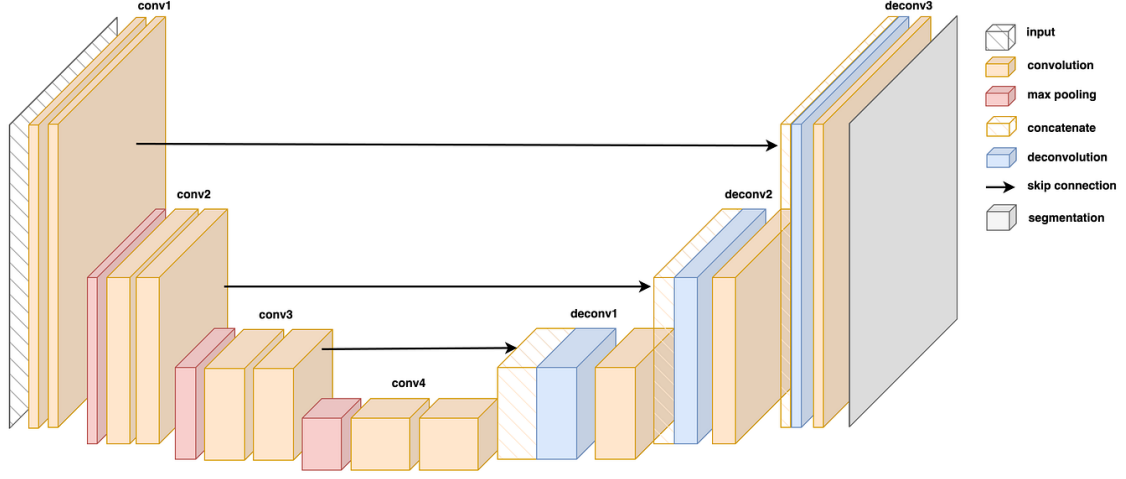


Figure 2.4: U-Net architecture.

Mathematically, let X_l denote the feature map at layer l of the encoder and Y_l the corresponding feature map in the decoder. In a standard encoder-decoder architecture, Y_l depends solely on the upsampled output of the previous decoder layer, such that $Y_l = f(\text{Upsample}(Y_{l-1}))$. In contrast, in the *U-Net* architecture [15], the input to each decoder layer is formed by concatenating the upsampled decoder features with the corresponding encoder features along the channel dimension, yielding $Y_l = f(\text{Concat}(\text{Upsample}(Y_{l-1}), X_l))$, where $\text{Concat}(\cdot)$ denotes channel-wise concatenation and $f(\cdot)$ represents the convolutional operations applied at that layer.

The skip connections allow the direct flow of high-resolution spatial information to bypass the bottleneck. Such a configuration allows the network to leverage the ability to combine the semantic contextual information learned in deeper layers with the textural features learned in the shallower layers. Thus, the network is able to achieve improved precision in the localization of the structures. Such precision is significant in the case of aorta segmentation since the accuracy in the boundary definition directly influences the accuracy in the computation of metrics such as the diameter and area.

3D U-Net

While the inspiration for the original *U-Net* architecture came from 2D biomedical images, structures such as the aorta have a 3D nature. By examining a CT volume on a slice-by-slice basis using a 2D network, there is a possibility of discontinuity along the longitudinal axis of the aorta due to a lack of contextual information regarding the continuity of a 3D volume. This might lead to "staircase" artifacts within a 3D reconstruction, which can affect the computation of centerline-based metrics.

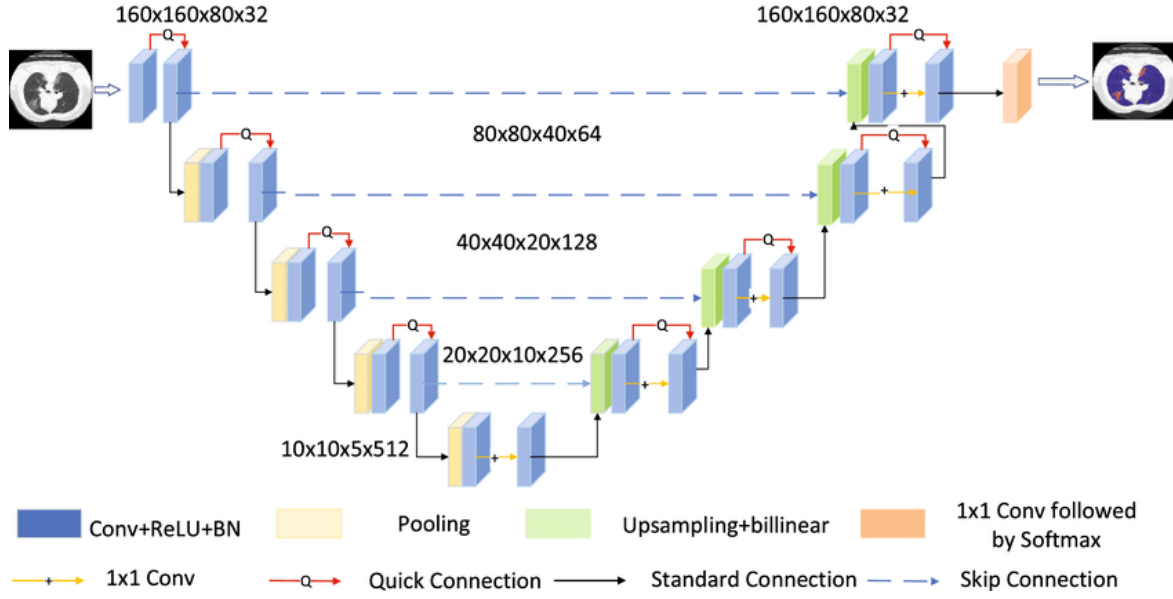


Figure 2.5: 3D U-Net architecture.

To overcome this, a 3D variant of *U-Net* was introduced. Essentially, in *3D U-Net*, all the 2D operations have been replaced by their 3D counterparts. In this network, 3D convolution operations, 3D max pooling, and 3D upsampling are used to process the patches and obtain information from all three dimensions simultaneously. As a result, the segmentation masks produced will have spatial consistency in the z-axis due to the use of inter-slice information.

However, there are significant computational challenges when moving to 3D architectures. This is because the memory requirements for gradients as well as feature maps grow cubically with the size of the input. Despite the computational costs of 3D networks, they are the only option for medical segmentation problems, where maintaining the topology of the organ as well as its smooth surface, is a requisite step towards further analysis.

2.2.4 Vision Transformers

In order to overcome the limitations of CNNs in representing long-distance dependencies, computer vision has seen a rise in models based on *Transformers*. *Transformers* are models that were first developed for natural language processing tasks and utilize an attention mechanism to weigh the influence of different parts of the input data on each other. When applied to vision tasks, they are known as ViTs [14].

Standard ViTs, unlike CNNs, receive the image as a sequence of patches. The image is divided into non-overlapping patches, which are further flattened and linearly projected to get feature embeddings. To capture the spatial information, which is present in CNNs but absent in the permutation-invariant architecture of *Transformers*, learnable position embeddings are added to these feature embeddings.

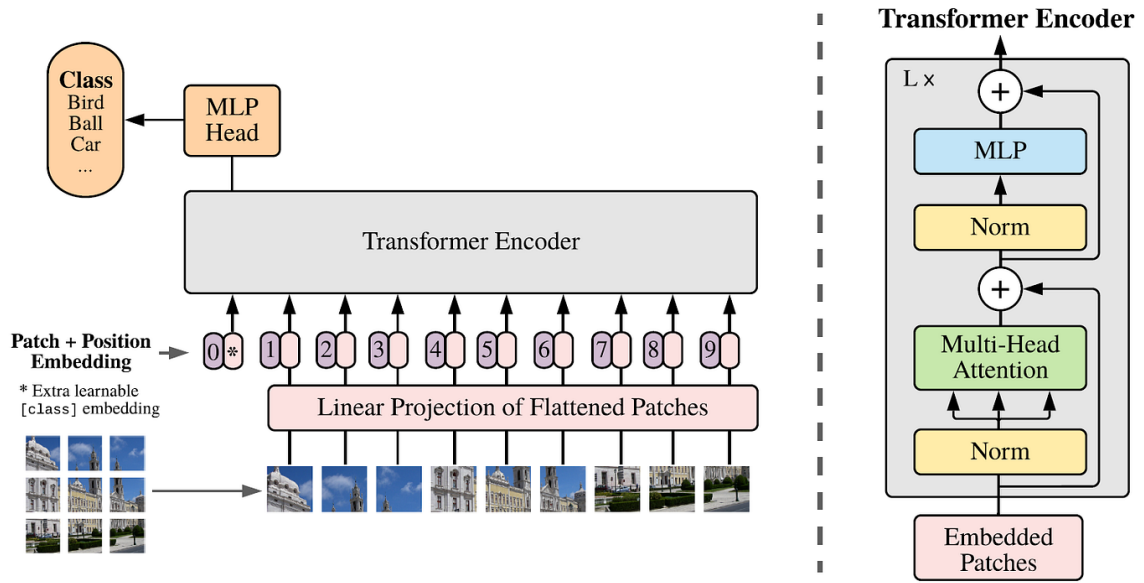


Figure 2.6: Vision Transformer architecture.

Self-Attention Mechanism

The central part of the *Transformer* architecture is the multi-head self-attention module. Self-attention allows the network to estimate the relevance of every image patch relative to every other image patch, regardless of the spatial distance between them. This allows the network to capture different anatomical relationships. For example, the network is capable of establishing the relationship between the geometry of the ascending aorta and the geometry of the descending aorta in the same volume scan, facilitating a more coherent segmentation of the entire vessel tree.

In mathematical terms, the self-attention mechanism learns to project the input features into three vectors, namely the *Key*, *Value*, and *Query*. Attention scores are computed as the dot product of the *Key* and *Query* vectors, controlling the extent of the focus placed on different parts of the input. This score is then used as a weight to the *Value* vector, generating an output that considers the global context of the input. While the benefit of the global context is considerable, the complexity of the self-attention mechanism grows quadratically with the number of tokens and becomes an obstacle when scaled to the high-resolution 3D volumes in the medical imaging context.

Hierarchical Transformers and Shifted Windows

In order to counter the computation complexity of the original ViTs [14] but still leverage the benefits of attention-based models, the concept of hierarchical models including *Swin Transformers* [16] was introduced. The *Swin Transformer* model incorporates a hierarchical model similar to that of CNNs but uses window-based self-attention instead of convolution.

In this approach, the input volume is divided into a series of small non-overlapping local

windows. The self-attention is calculated only within the local windows, thereby reducing the computational complexity from quadratic to linear with respect to the size of the image. Nonetheless, solely depending on local windows would again impose the receptive field limitation of CNNs. This challenge is overcome by the use of a shifted window partitioning approach, between consecutive layers.

This hybrid approach successfully integrates the efficiency of local processing with the global modeling power of *Transformers*. For medical image segmentation tasks, models that incorporate this approach like *SwinUNETR* [17], have achieved state-of-the-art results. They are able to extract the local details needed to define the boundaries of the aortic wall with a global view of the topology of the entire vessel, which is important in analyzing features and geometry.

2.2.5 Hybrid and Transformer-based Architectures

Recent studies have demonstrated that ViTs [14] have strong capabilities in modeling long-range dependencies through self-attention, even though they generally do not possess some of the inductive biases characteristic of Convolutional Neural Networks, namely translation invariance and locality. In the task of segmenting the aorta, both local and global features of the image have been found to be vital. On the one hand, the local features help in accurately specifying the boundaries of the aorta wall, which in turn can help in the precise measurement of the diameter of the aorta. On the other hand, the global features help in the differentiation of the aorta from other tubular structures of similar radiodensity, like the vena cava and the esophagus, and maintaining the continuity of the volume of the aorta in the longitudinal direction. For this reason, hybrid architectures were proposed, incorporating *Transformers* into an existing U-shaped encoder-decoder model.

UNETR

A common hybridization paradigm is to replace the convolutional encoder of the standard *U-Net* with a *Transformer* backbone, giving rise to architectures such as *UNETR* [18]. In this configuration, the 3D input volume is split into a sequence of uniformly sized and non-overlapping patches and projected to a linear embedding space. Subsequently, these projected patches are fed through a sequence of *Transformer* layers with global self-attention to extract semantic features of anatomy. The features from different *Transformer* encoder layers are rearranged and upsampled to integrate with the convolutional decoder via skip connections. While this configuration is efficient in capturing global anatomical context for the aorta and using the convolutional decoder to improve the resolution of the segmentation mask, it is also computationally expensive for *Transformer* architectures to process high-resolution images and 3D data, due to the quadratic complexity of self-attention mechanisms with respect to the sequence length.

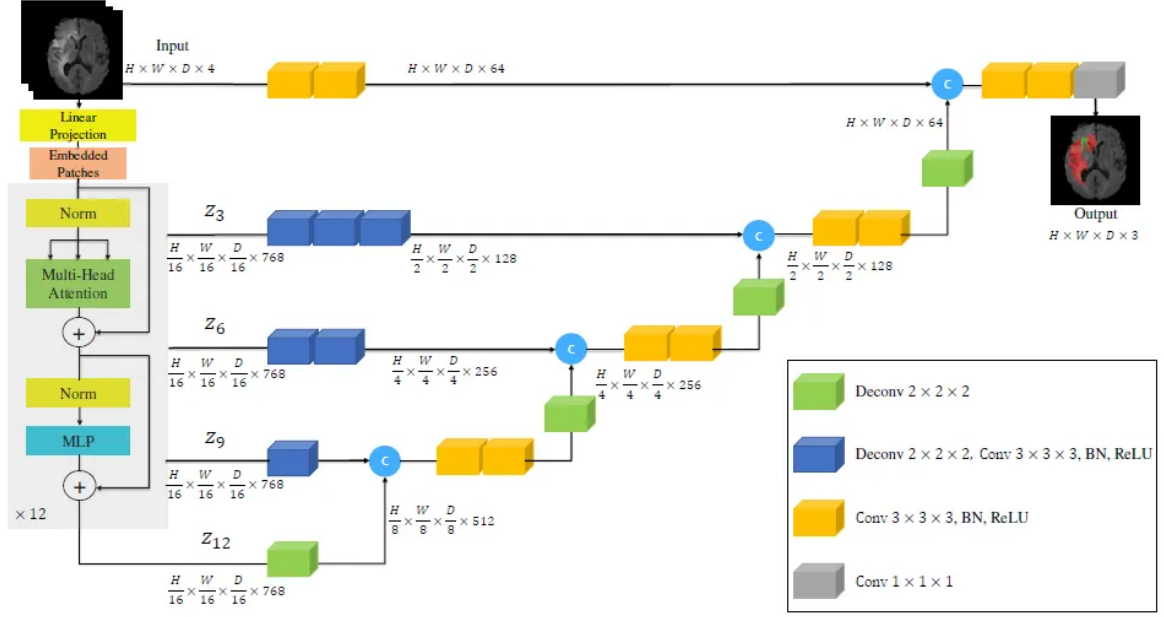


Figure 2.7: UNETR architecture.

SwinUNETR

To reduce the computation cost of *Transformers* but retain the advantage of the U-shaped structure, the *SwinUNETR* was proposed. This architecture is based on the hierarchical structure of the *Swin Transformer* [16] but is applied in the context of medical image segmentation. While regular *Transformers* calculate attention in the whole sequence of the input image, in the case of *SwinUNETR*, self-attention computation is only applied locally within non-overlapping windows. This local computation of attention has reduced the computation cost from quadratic to linear in relation to the size of the images.

Another important feature of the *SwinUNETR* encoder is the proposed Shifted Window mechanism. Although the proposed window-based self-attention calculation reduces computational inefficiency, it also restricts the information within the window boundaries. To overcome this limitation, the *SwinUNETR* architecture utilizes two different partitioning schemes for the windows. In the first layer, the traditional window partitioning scheme is used. In the second layer, the window partitioning scheme is shifted, thus shifting the windows from the previous layer. This shift bridges the boundaries of the previous windows, thus enabling a progressive expansion of the receptive field with the depth of the network.

In the case of aortic analysis, the proposed *SwinUNETR* architecture provides a significant advantage. The hierarchical structure of the encoder provides feature maps of varying resolutions, which are connected to a residual convolutional decoder via skip connections. The *Transformer*-based encoder can efficiently model the topological continuity of the aorta in the image volume, thus reducing false negatives in tortuous regions or aneurysmal regions where the geometry deviates from the norm. At the same time, the convolutional decoder and skip connections ensure the preservation of fine spatial details, thus enabling the ex-

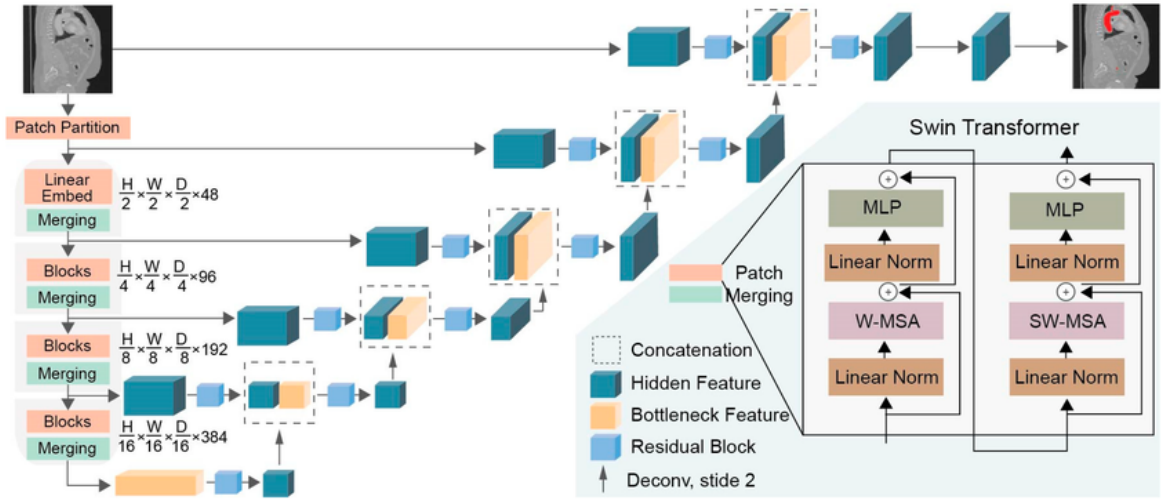


Figure 2.8: SwinUNETR architecture.

traction of smooth surface meshes for the subsequent geometric analysis.

2.3 State of the Art in Medical Segmentation

2.3.1 Evolution of Computational Methods for Aortic Analysis

The segmentation of the aorta and the associated vascular tree is a fundamental prerequisite for the assessment of various cardiovascular pathologies, including aneurysms and dissections. In the past, the delineation of the aortic wall from the data provided by the CT scans and MRI scans was primarily reliant on manual or semi-automatic annotations. Although the manual delineation is often used as the ground truth for validation purposes, this is often associated with a number of drawbacks, including a high degree of labor. As a result, the research community has increasingly moved towards fully automatic computational solutions.

Traditional and Deformable Model Approaches

The traditional approaches for aortic segmentation primarily involve model-based approaches and tracking algorithms. The Hough transform was used for detection of circular cross-sections of the aorta from axial slices of images, which are usually used as an initial step toward active contour models as well as level set approaches. These deformable model approaches are usually characterized by energy minimization. The energy function is a combination of the internal forces, which are often responsible for the smoothness of the model, and the external forces, which are often derived from the gradients of the image. These approaches are usually effective for high-contrast CTA images, where the aorta lumen is well defined, but they have failed to deliver satisfactory results in cases of ADs, where the irregular nature of the false lumen, as well as the presence of intimal flaps, may cause significant topological changes, which are difficult to model using energy minimization approaches [19]. Moreover, non-contrast CT images are characterized by significant complexity, where dif-

ferentiation of blood pool from surrounding tissues is difficult, which necessitates complex multi-atlas registration approaches for achieving acceptable accuracy of aortic segmentation [20].

Deep Learning and Convolutional Neural Networks

The emergence of deep learning and CNNs has marked a paradigm shift in the analysis of medical images. CNNs eliminate the need for manual feature extraction, as they are able to automatically learn from input data. The *U-Net* architecture [15], characterized by its symmetric encoder-decoder structure and skip connections, has emerged as a standard solution for semantic segmentation tasks. Recent studies have demonstrated the effectiveness of CNNs for the analysis of the heterogeneity of aortic pathologies. For example, *Mavridis et al.* [21] presented a comparative analysis of the effectiveness of the *U-Net* [15], *SegResNet* [22], *UNETR* [18], and *SwinUNETR* [17] architectures for the purpose of segmenting AAAs. The authors concluded that while the *Transformer* architectures have the advantage of considering global context information, the established *U-Net* and the variant of the *SegResNet* network often yielded higher segmentation accuracy and computational efficiency on small datasets.

he application of CNNs is not limited to the lumen segmentation but also to the characterization of different pathologies. In the case of AAA, recent frameworks have used a combination of segmentation and active learning strategies, resulting in high accuracy while reducing manual effort, while also achieving high accuracy in the measurement of morphological parameters such as maximum diameter and volume [23]. deep learning algorithms have also been validated for the automatic detection of aortic dissections, as well as quantification of thoracic aortic enlargement, with a strong correlation with manual measurements carried out by expert radiologists [24].

Transformers and Hybrid Architectures

In spite of the success of CNNs, they also have some inherent limitations in terms of the receptive field, which limits the ability of the model to capture the long-range dependencies in a volumetric scan. To address these limitations, recent studies have explored the application of *Vision Transformers* in medical image segmentation tasks. The *Transformers* use self-attention to weigh the importance of different parts of the data, which allows the model to better comprehend the overall context of the image, rather than the local context of CNNs. However, the application of *Transformers* is not without limitations, as some studies suggest that the application of *Transformers* requires a much larger dataset to achieve better results than CNNs, which may also be computationally more expensive [21]. In the case of intravascular imaging modalities such as Intravascular Ultrasound (IVUS), it is noted that *Transformers* are also being explored in order to effectively differentiate vessel wall boundaries, though bias and interpretability are challenges in these advanced systems [25].

Challenges in Pathological Segmentation

The challenges in the state of the art in this field include the accurate segmentation of the aortic vessel tree in the presence of severe pathology. In ADs, in which the aortic wall is divided into true and false lumen, is a challenging segmentation problem. The contrast uptake in this case is difficult, and the intimal flap is very thin. According to *Pepe et al.* [19], in spite of several methods being applied to the healthy aorta and simple aneurysms, the accurate segmentation of the entire vessel tree, including the dissection membrane and false lumen, is a challenging problem, and specialized approaches must be applied. The ability of these models in effectively generalizing across multi-center datasets is also a critical area of investigation, and it is noted that domain shifts in these scanners would affect the performance of these models significantly [26].

2.3.2 MICCAI SEG.A. 2023 Challenge

In order to address the challenges associated with anatomical variations and heterogeneity of imaging acquisition protocols, the field of medical image computing has increasingly used standardized benchmarking competitions. The Segmentation of the Aorta (SEG.A.) 2023 challenge, which took place in conjunction with the Medical Image Computing and Computer Assisted Intervention (MICCAI) conference, aimed at assessing the accuracy of algorithms used in segmentation, modeling, and meshing of the Aortic Vessel Tree (AVT). The main goal of this challenge was to move beyond simple diameter measurements towards fully automated 3D reconstructions, which are not only accurate but also useful for a range of potential applications, including diagnosis, surgical planning, or fluid dynamics simulations [27].

Dataset Characteristics

The challenge used the multicenter AVT dataset, which was designed to represent a heterogeneous population. The dataset consists of 56 CTA scans, with data obtained from three different institutions, including the KiTS19 Grand Challenge, the RIDER Lung CT dataset, and clinical data from Dongyang Hospital. The dataset includes a range of variations in terms of imaging acquisition parameters, including different manufacturers, resolutions, slice thickness, and reconstruction kernels, as described in table 2.1. The dataset includes a range of healthy as well as pathological cases, including AAAs and ADs. The ground truth segmentations include the entire AVT, including the ascending aorta, aortic arch, descending thoracic aorta, abdominal aorta, and common iliac arteries [28].

Challenge Tasks and Evaluation

The MICCAI SEG.A. 2023 challenge had three tasks. The first task was volumetric segmentation of the AVT. Participants had to submit a containerized algorithm that was tested using a hidden test set. The evaluation criteria were the DSC to measure volume overlap, and the 95th percentile of the Hausdorff Distance (HD95) to measure the accuracy of the boundaries. A significant factor was the algorithm’s robustness to different inputs, which was tested by using the original test cases as well as their augmentations [27]. The other tasks were the

Image Information	KiTS	RIDER	Dongyang
x/y resolution	512 x 512	512 x 512	512 x 666
Axial slices	94/146/1059	260/1008/1140	122/149/251
Slice thickness	0.5/5/5 mm	0.625/0.625/2.5 mm	2/3/3 mm
Pathologies	None	AD, AAA	None
Vessel Tree Volume	55.8/284.5/464.4 ml	176.6/354.1/614.0 ml	126.0/254.9/488.1 ml
Segmentation Times	30/118/30.5 min.	27/422/80 min.	12/35/19.5 min.
Number of Cases	20	18	18

Table 2.1: Image information of the aortic vessel trees. Number of axial slices, segmentation times, slice thickness and vessel tree volume are given as: min/median/max.

generation of a surface mesh for visualization and the generation of a volumetric mesh for simulation. This highlighted the importance of the segmentation being not only voxel-wise accurate but also topologically correct.

Leading Approaches

The results of the challenge highlighted the dominance of deep learning approaches in comparison to the traditional approaches. The winning approach for the first task of the challenge was presented by *Myronenko et al.* [29]. They used an ensemble of 15 models of the *SegResNet* architecture [22], with the models being trained using the 5-fold cross-validation approach. The approach achieved an average DSC score of 0.920 and an HD95 of 6.013 mm on the final hidden test set. The most important part of the winning approach was the use of the *Auto3DSeg* pipeline, which is part of the MONAI framework [30], and allows for the automatic configuration of the hyperparameters and extensive data augmentation to deal with the heterogeneity of the source data. The other best-performing methods also used variants of the *U-Net* architecture [15], showing that encoder-decoder architectures remain the state of the art for this specific segmentation problem.

2.3.3 MONAI Framework

The advancement of sophisticated segmentation pipelines in the field of medical imaging has called for the development of specialized computational tools with the ability to efficiently manage high-dimensional volumetric data, as well as domain-specific data formats such as NIfTI and DICOM. MONAI [30] is an increasingly popular, open-source software framework specifically tailored for the implementation of deep learning-based applications in the field of medicine. MONAI is built on the PyTorch library and offers a comprehensive range of modular tools optimized for the implementation of medical image analysis tasks.

One of the primary characteristics of the MONAI framework [30] is the availability of state-of-the-art neural network architectures that are optimized for the implementation of various tasks related to the analysis of medical images. This includes the implementation of well-established convolutional neural networks such as the *3D U-Net* [31] and *SegResNet* [22], as well as emerging transformer-based architectures such as *UNETR* [18] and *SwinUNETR* [17]. Recently, several comparative studies have utilized the MONAI library to implement

the above architectures for the evaluation of their performance on standardized tasks. For example, *Mavridis et al.* [21] utilized the MONAI framework to evaluate the performance of both convolutional and transformer-based architectures for the segmentation of the abdominal aorta, while *Kim et al.* [23] utilized the MONAI implementation of the *UNETR* [18] and *SwinUNETR* [17] architectures to extract morphological features for aneurysm measurement.

The *Auto3DSeg* pipeline is a feature within the MONAI framework that is a significant advancement towards the automation of the configuration of image segmentation tasks. The pipeline is capable of analyzing the statistical properties of a given dataset and generating heuristic-based hyperparameters that are required for the implementation of the segmentation task. The effectiveness of the *Auto3DSeg* pipeline was demonstrated in the MICCAI SEG.A. 2023 challenge [27], where the winning entry utilized the pipeline to implement an ensemble of *SegResNet* architectures [29]. The framework is capable of autonomously handling critical preprocessing steps like intensity normalization and resampling, thereby enabling the robust handling of the heterogeneity present within the multi-center dataset.

2.4 Geometric Analysis and Feature Extraction

The quantitative evaluation of aortic pathologies goes beyond the primary segmentation of anatomical structures. While semantic segmentation methods provide voxel-wise classification of medical images, clinical decision-making is based on specific morphological and geometric features such as the diameter, volume, tortuosity, and local curvature of vessels. Geometric analysis and feature extraction form the bridge between the volumetric masks obtained from deep learning methods and the quantitative features used by radiologists and surgeons. In recent years, the construction of automated methods for the quantitative evaluation of these features has been reported in the literature. The ultimate aim is to minimize the variability between observers and the time required for the evaluation process. The general workflow for geometric analysis and feature extraction consists of a series of steps, such as 3D surface mesh generation, vessel centerline extraction, cross-sectional measurements, and advanced shape feature calculation.

2.4.1 Centerline Extraction and Skeletonization

The centerline of a vascular structure is considered a fundamental geometric abstraction that follows the central axis of the vessel lumen along its longitudinal axis. It is considered the primary geometric reference for subsequent morphological measurements because standard measurements require strict perpendicularity between the cross-sections and the local vessel axis. In the absence of an accurate centerline, the measurements of the maximum diameter can be significantly overestimated due to oblique cutting planes.

Traditionally, centerline extraction algorithms have been classified into three main types: morphology-based, distance transformation-based, and model evolution-based approaches [32]. Morphology-based approaches, also known as skeletonization, work by iteratively thinning the segmentation mask. Another way of defining the centerline is through the distance mapping approach. This is achieved by calculating the distance from the boundary field within the segmented lumen. The centerline is then determined by the gradient of the distance map. This ensures that the path is always as far as possible from the boundary. In a recent automated algorithm for the characterization of AAAs developed by *Vila Delgado* [33], combined a Gaussian filter for the smoothing of the combined mask of the lumen and the ILT, followed by 3D skeletonization for the determination of the centerline of the vessel. The minimum spanning tree algorithm was used for the determination of the centerline of the vessel. The determined centerline was subsequently smoothed using a spline interpolation function to account for the discrete coordinates of the image voxels.

The 3D Voronoi diagram-based approach is an example of an algorithm that calculates paths that have the least total integral of the radius of the maximum inscribed spheres. The Voronoi diagram is effective in defining the medial surface of the object. The graph search algorithm is then used to define the shortest path between the endpoints. The Voronoi diagram is implemented in the open-source library VMTK, which is used to compute centerlines of branching tubular objects [34].

Recently, deep learning methods have also been investigated for the direct extraction of centerlines from image data or segmentation masks. For instance, *Liu et al.* [32] presented a dual-branch topology-aware *U-Net* (*DTUNet*), which simultaneously predicts the lumen segmentation and vessel centerline. In another study, *Epifanov et al.* [35] presented a multi-task network that used graph layers to directly predict the points of the vessel centerline with real-valued coordinates. In this way, the post-processing steps that are often necessary for voxel-wise skeletonization methods are avoided. In addition, reinforcement learning methods have also been investigated for the detection of anatomical landmarks, such as the aortic root or branch bifurcations, which are then used to guide the extraction of the vessel centerlines [24].

2.4.2 Cross-Sectional and Diameter Assessment

With the smooth centerline now in place, the three-dimensional volume of the aorta can now be analyzed through a series of two-dimensional cross-sections. This is accomplished by discretizing the centerline into points at regular intervals (for instance, every 1 mm or every 5 mm). The tangent vector is then calculated at each of these points. The cross-section plane is then determined by the orthogonal complement of the tangent vector. The 3D binary mask is resampled along this plane to produce a true cross-section of the vessel.

Automated diameter and area calculations are heavily dependent upon this process. As

shown in *Gamechi et al.* [20], automated calculations can determine the cross-sectional average diameter of the aorta at regular intervals relative to specific anatomical landmarks. The maximum diameter can be calculated as the largest distance between contour points. Alternatively, algorithms such as maximal inscribed sphere calculate diameter directly in three-dimensional space by determining the largest sphere that fits inside the vessel boundary at a particular point along the centerline [34].

Besides the regular diameter measurements, cross-sectional analysis can also be used to compute shape descriptors such as circularity, eccentricity, and ellipticity. Circularity is defined as the measure of the similarity of the cross-sectional shape to a perfect circle. A small value of the circularity index can be an indication of an irregular aneurysm shape, the presence of a thrombus, or extensive calcifications, which can be associated with weakened arterial walls.

2.4.3 Surface Mesh Generation and Volumetric Analysis

To perform three-dimensional geometric analysis of the aorta, the segmented masks are usually transformed into surface meshes. The Marching Cubes algorithm is commonly used for isosurface reconstruction from volumetric data. This algorithm results in a mesh comprising vertices and triangular faces. However, due to the anisotropic nature of the CT data, the binary voxel masks may exhibit a staircase effect, necessitating the use of smoothing algorithms to produce realistic representations of the aortic wall.

To perform more advanced meshing for hemodynamic simulations, *Bosnjak et al.* [36] developed a semi-automatic method for creating a block-structured hexahedral mesh for aortic dissections. This method involves creating a topological skeleton which provides a basis for creating a block-structured mesh using transfinite maps.

Once the surface meshes and voxel matrices are defined, the volumetric parameters can be quantified with great accuracy. The total vessel volume, lumen volume, and the volume of thrombus or calcifications can be computed by integrating the voxel sizes in the specific segmented regions.

2.4.4 Tortuosity, Curvature, and Advanced Shape Descriptors

In addition to these basic cross-sectional dimensions, the trajectory and global configuration of the aorta are also crucial determinants of hemodynamic efficiency and surgical intervention. The parameters of tortuosity and curvature are geometric parameters that describe the complexity of the vessel's trajectory. These parameters are especially important in endovascular procedures, where extreme vessel curvature may compromise catheter and stent placement.

Tortuosity is typically calculated using a distance metric that is defined as the ratio of the actual centerline length to the Euclidean distance between vessel endpoints. A value of 1.0 signifies a perfectly linear path, and increasing value signifies increasing degrees of twisting. *Canals et al.* [37] proved the effectiveness of a fully automated pipeline to extract tortuosity feature from the geometry of supra-aortic vessels. The geometric feature was extracted and validated, and relative length and departure angles were compared to manual measurements. The study proved that a fully automated pipeline could be used to obtain objective data to assess catheter access in acute stroke cases, a notion that could be directly applied to aorta access for aneurysm repair.

The local curvature parameter provides a finer assessment of vessel geometry. The mathematical definition of local curvature is given by the reciprocal of the radius of the osculating circle. Local curvature is calculated from derivatives of smoothed centerline coordinates. High curvature is clinically important, especially in areas where there is a high likelihood of abnormal blood flow and high shear stress, contributing to vascular diseases.

Advanced shape descriptors further expand the capabilities of geometric analysis by considering the vessel as a whole 3D object rather than just a tube. Statistical shape modeling is a tool that can be used for the analysis of the global variation of the vessel shapes in a population of patients. *Bruse et al.* [38] developed a non-parametric framework for statistical shape modeling that eliminates the need for manual landmarking. In their study, the authors considered 3D shapes as “currents” (mathematical distributions of shape features). They calculated the mean anatomical template and studied how the individual patient shapes deformed from the mean. This allowed the researchers to identify 3D vessel shape biomarkers, such as specific arch morphologies (for example, “gothic” arches), which correlated with functional parameters such as ejection fraction in patients with repaired aortic coarctation.

Geronzi et al. [39] used a geometric decomposition technique to extract local and global vessel shape features for the prediction of the growth of the AAA. The authors used the ratio of the vessel diameter to the centerline, the ratio of the external to internal curvature lengths, and other descriptors. The results of the study showed that the integration of the global vessel shape modes into regression models can potentially exceed the performance of models using only local measurements, suggesting that the topological configuration of the aneurysm has an independent effect on its progression.

2.5 Summary and Discussion

The literature reviewed in this chapter shows that there is a shift in the field of medical image analysis from traditional, labor-intensive methods to more automated and data-driven methods, enabled by the recent progress in deep learning. The clinical evaluation of aortic

diseases, such as aneurysms and dissections, has traditionally used 2D parameters such as the maximum diameter. However, recent studies have emphasized that such unidimensional parameters are not adequate to capture the complex 3D changes in aortic morphology associated with disease progression and rupture risk.

In the field of semantic segmentation, CNN models, such as the popular *U-Net* and its 3D variants, have provided a solid foundation for anatomical segmentation tasks. Recently, the introduction of *Transformer* modules, such as *SwinUNETR*, has improved the ability of these models to capture complex dependencies, which is a key issue in the segmentation of large, complex structures such as the aorta. The recent results from the MICCAI SEG.A. 2023 challenge have shown that although high overlap can be achieved in the segmentation task, the key issue that persists is the difficulty in maintaining topological accuracy in the segmentation results, especially in the segmentation of pathological structures such as ILT and dissections. The ability to generalize these models to multi-center datasets with diverse imaging protocols is a significant challenge.

Subsequent to segmentation, the extraction of geometric features acts as the interface between voxel data and clinical understanding. The derivation of vessel centerlines is of prime importance, as this enables the computation of accurate dimensions. Although traditional methods based on thinning algorithms are still being used, recent studies indicate that the prediction of vessel centerlines using deep learning can provide improved robustness against artifacts and discontinuities. Moreover, the generation of accurate surface meshes, can aid in advanced volumetric analysis and is essential for the computation of fluid dynamics simulations.

Despite these technological advancements, a significant gap remains in the integration of these individual components into cohesive, fully automated clinical workflows. The current state-of-the-art methods reported in the literature are mostly focused on addressing specific components of the system, either segmentation or centerline prediction, and often require manual intervention for initialization and landmark detection. Furthermore, although advanced shape descriptors are being recognized as promising biomarkers for risk stratification, they are still not being computed in clinical practice due to the lack of standardized and user-friendly computation tools.

The research in this dissertation aims to bridge this gap and create an end-to-end solution using state-of-the-art architectures for segmentation and robust algorithms for geometric analysis. The proposed solution can help in automating the entire clinical analysis pipeline from the initial CT scan data to the generation of morphological reports, which can aid clinicians in decision-making regarding the treatment of aortic pathologies.

Chapter 3

Proposed Method

3.1 Dataset Description

The dataset used in the present work is the multicenter AVT dataset, which is the basis for the MICCAI SEG.A. 2023 Challenge. The dataset comprises 56 CTA scans, which were collected from three different centers: the KiTS19 Grand Challenge, the RIDER Lung CT, and Dongyang Hospital.

The data exhibits significant heterogeneity regarding image acquisition parameters. As detailed in table 2.1, variations exist in scanner manufacturers, spatial resolutions, slice thicknesses, and reconstruction kernels. The distribution of slice thickness across the different scan sites is illustrated in figure 3.1, while figure 3.2 depicts the pixel-level histograms, indicating that scans from different sites do not share identical intensity distributions.

The dataset contains various physiological conditions, including normal anatomy, AAAs and ADs. The ground truth segmentations contains the entire vessel tree, including the ascending aorta, the aortic arch, the descending thoracic aorta, the abdominal aorta, and the common iliac arteries. Figure 3.3 provides a visual example of the AVT data structure.

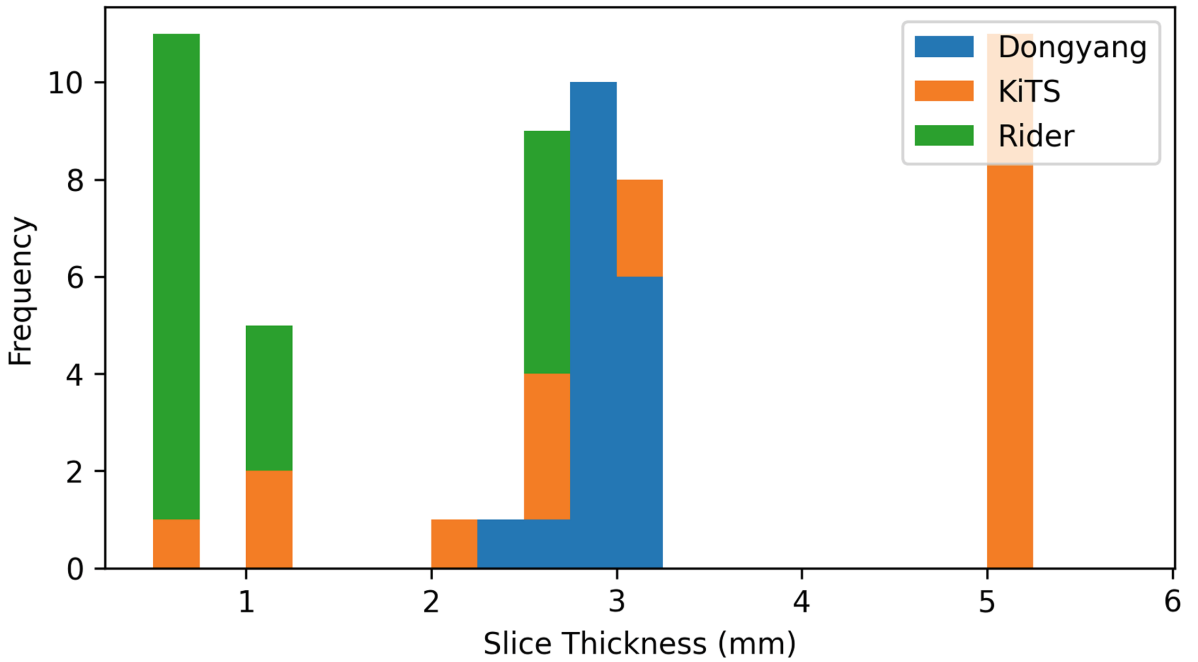


Figure 3.1: Distribution of slice thickness for each CT scan. Scans are colored by site.

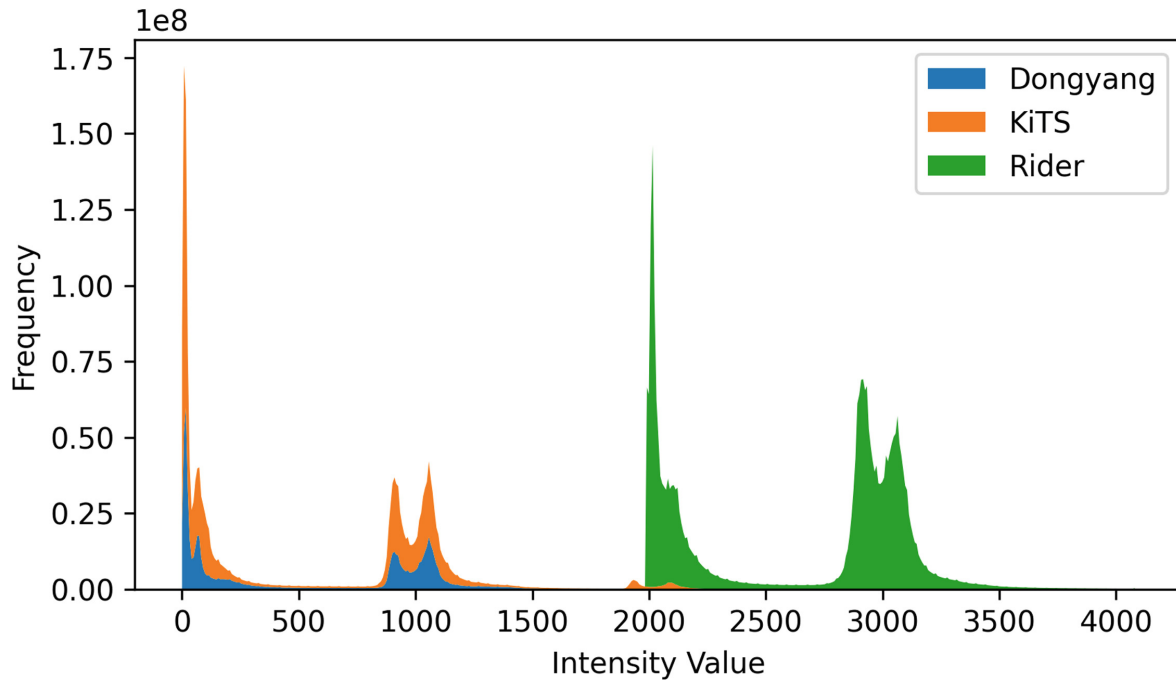


Figure 3.2: Pixel level histogram. Scans are colored by site. This shows that scans from different sites do not have similar distribution.

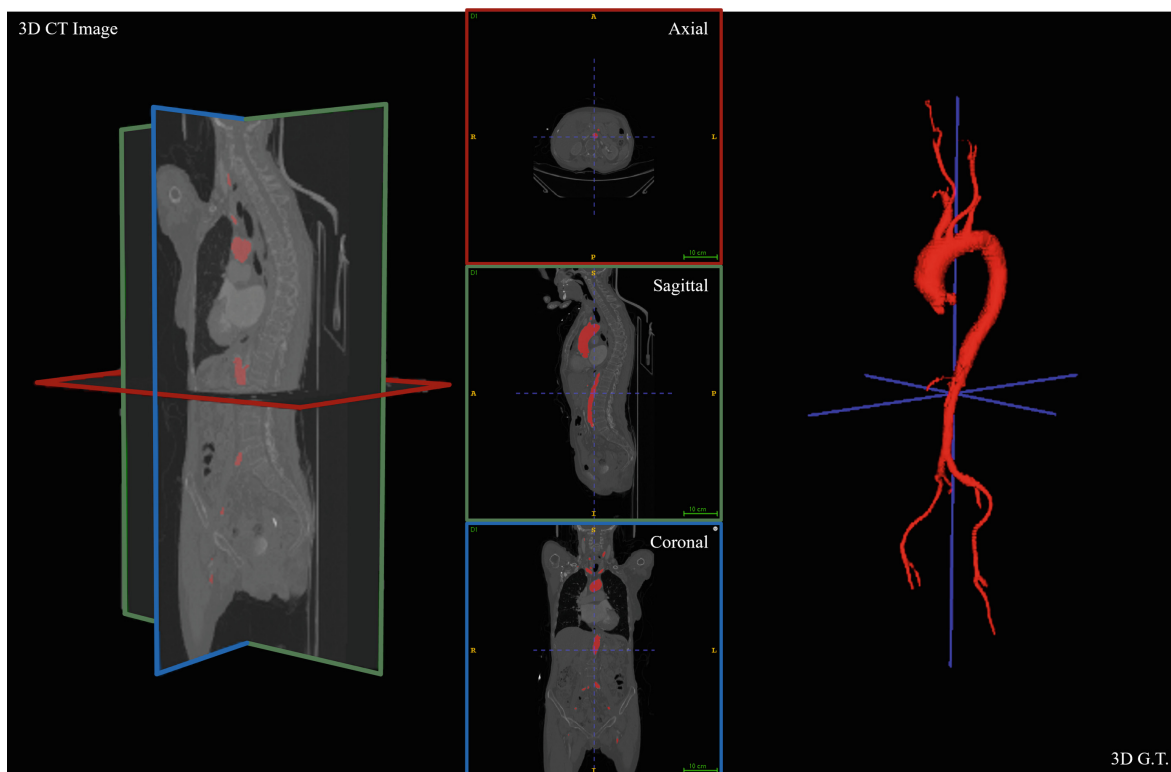


Figure 3.3: Example of the AVT data.

3.2 System Overview

The primary objective of this method is to facilitate the clinical evaluation and longitudinal monitoring of aortic disease progression. This is achieved through the development of a

fully automated computational framework capable of generating surface meshes for the visualization of the aorta and the monitoring of quantitative geometric descriptors. A schematic overview of the proposed framework is presented in figure 3.4.

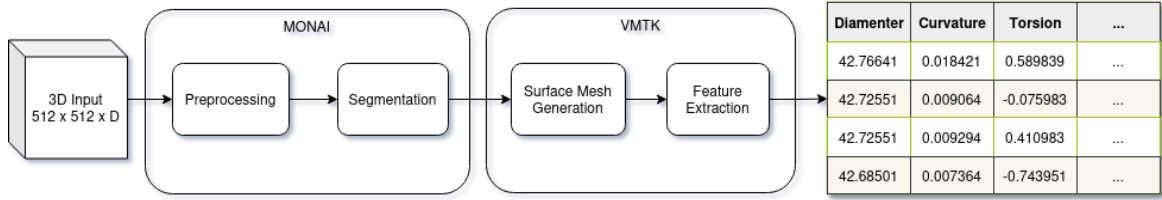


Figure 3.4: Diagram overview of the method's framework.

Preprocessing

The computational process begins with a series of preprocessing steps that are aimed at normalizing the input data. This includes a series of steps where the input data is reoriented to a standardized default anatomical position. This is followed by the normalization of the intensity values of the voxels in the input data. The input data is then resampled to a standardized voxel dimension of $1 \times 1 \times 1 \text{ mm}$ to ensure that the spatial resolutions of every aortic volume is consistent.

Segmentation

The segmentation process is carried out using the MONAI library. This library was selected based on the comprehensive range of modular tools optimized for medical image analysis tasks. The segmentation model is based on the *SwinUNETR* architecture. Although the existing literature shows that the *SegResNet* model might be more effective for smaller datasets, the model was able to learn the data effectively, achieving a validation DSC of 0.917 by the end of the training phase, as shown in figure 3.5, and an average of 0.93 on the test set.

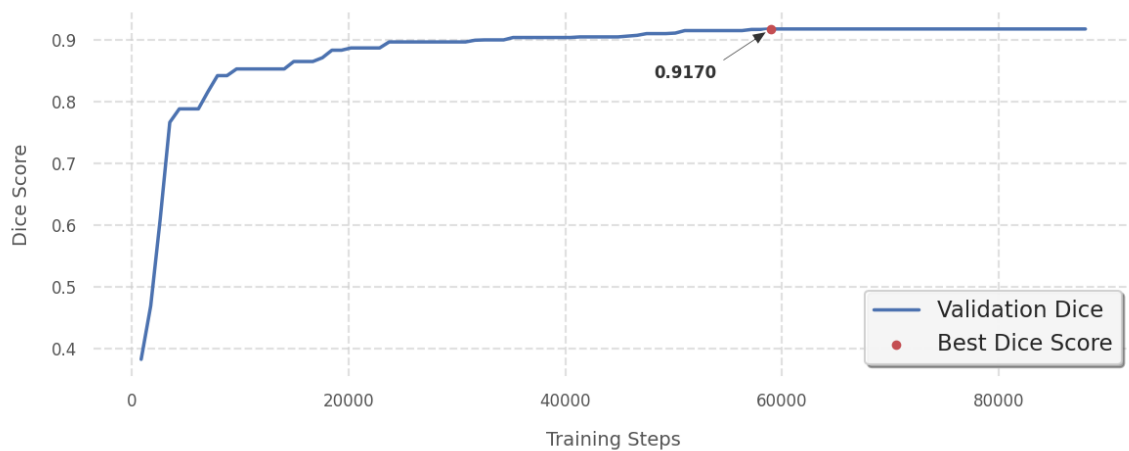


Figure 3.5: Validation Performance (Dice Score) during *SwinUNETR* segmentation training.

Feature Extraction

After segmentation, volumetric masks are processed into surface meshes using the VMTK. This toolkit can handle three-dimensional reconstruction and geometric analysis for image-based vascular modeling. Centerline extraction is performed using a Voronoi diagram–based method from the toolkit. Quantitative morphological parameters can be obtained using the surface mesh and its corresponding vessel centerline. Quantitative morphological parameters include vessel diameter (figure 3.6), curvature (figure 3.7), torsion (figure 3.8), and tortuosity (figure 3.9). These parameters have corresponding graphs showing the value of each parameter at certain points along the vessel centerline, including branch points. In the vessel diameter analysis, a gradual decrease in vessel diameter can be observed along the longitudinal axis, especially at branch points.

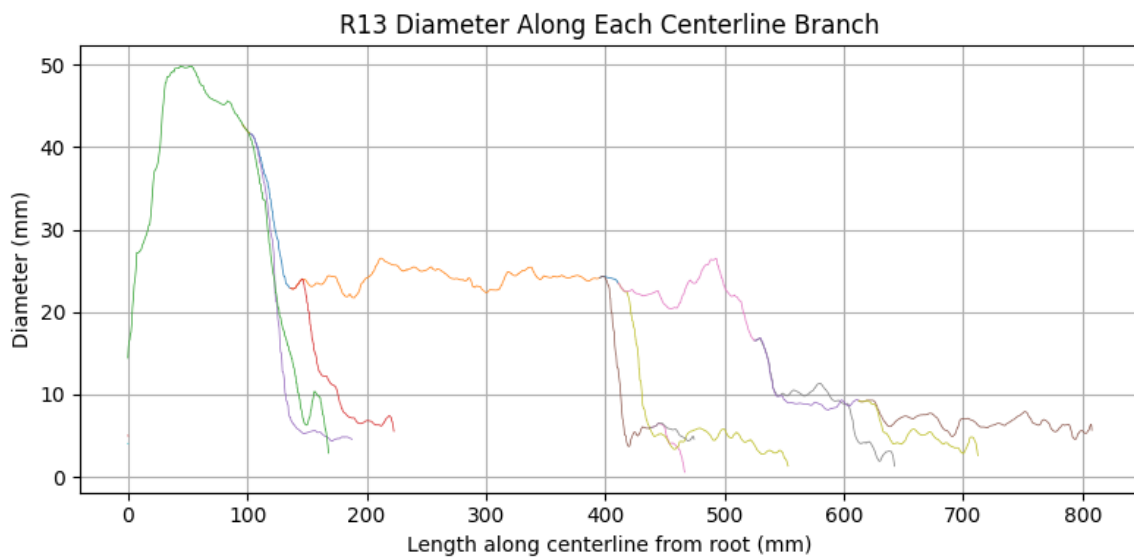


Figure 3.6: Aorta diameter measurement along the centerline of the vessel, and respective branches.

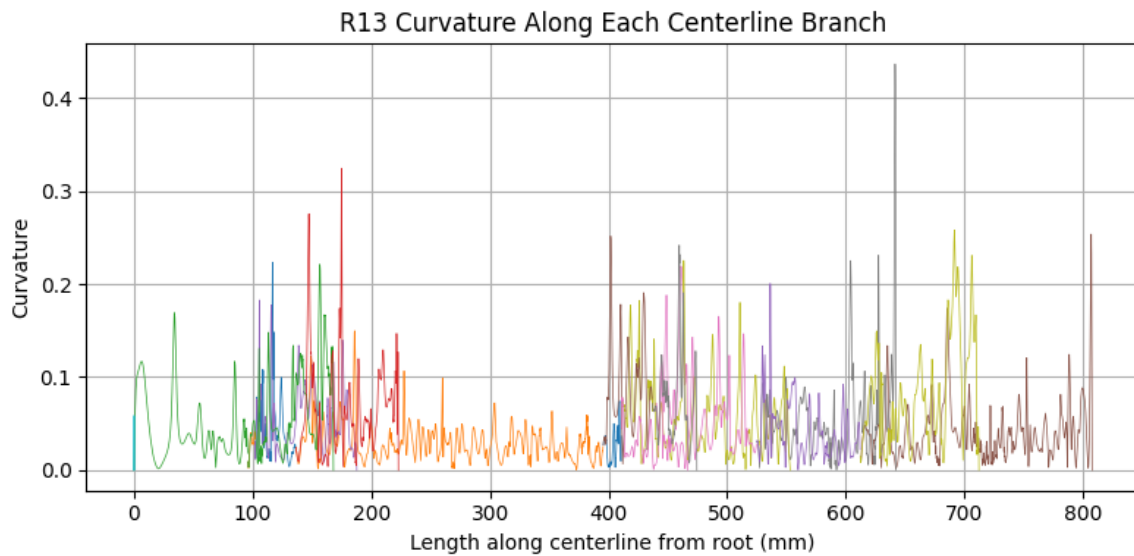


Figure 3.7: Aorta curvature measurement along the centerline of the vessel, and respective branches.

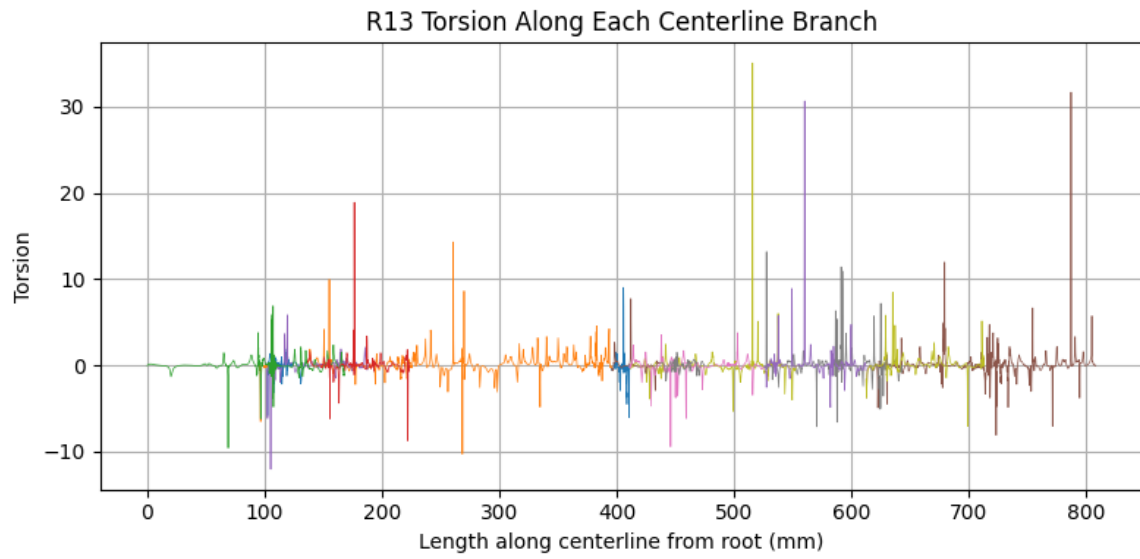


Figure 3.8: Aorta torsion measurement along the centerline of the vessel, and respective branches.

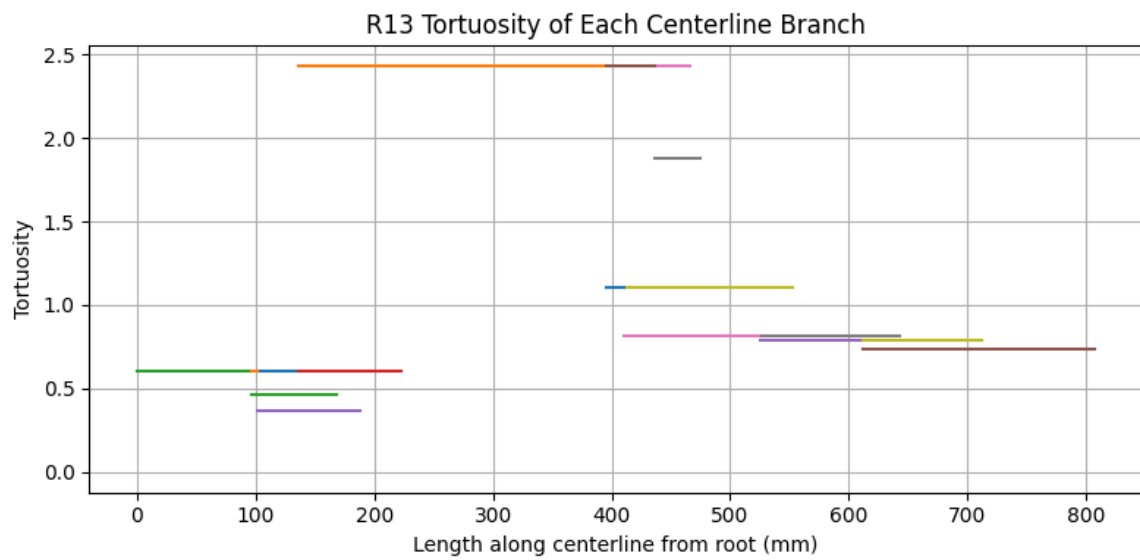


Figure 3.9: Aorta tortuosity measurement of each segment of the vessel. A new segment is created at every bifurcation. The respective tortuosity value of each segment is calculated based on the respective start and end point.

Chapter 4

Work Planification

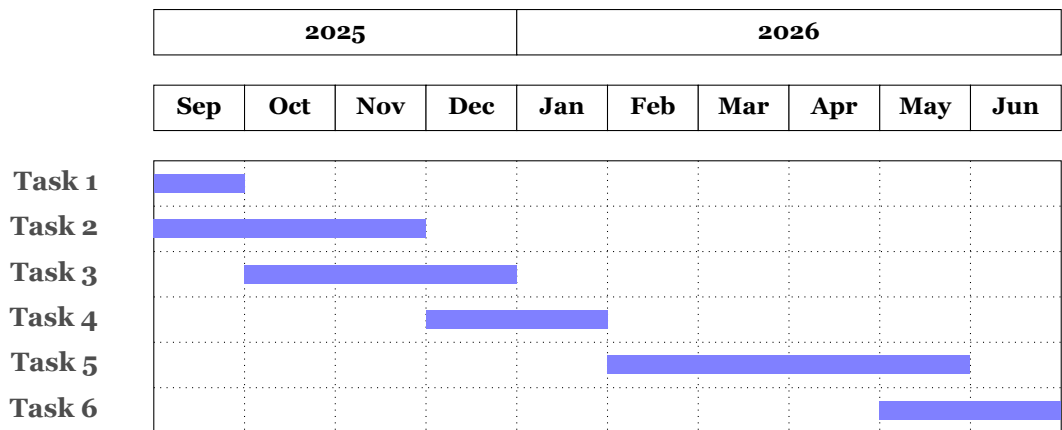


Figure 4.1: Gantt chart illustrating the development timeline and duration of the proposed tasks.

- **Task 1 - Dataset Research:** Concurrently with the literature review, research was conducted to identify a suitable dataset containing segmented aortic data. Due to privacy restrictions regarding medical data, the search prioritized public datasets that met the complexity requirements for deep learning applications.
- **Task 2 - Literature Review:** The review focused on medical image segmentation methodologies and feature extraction techniques relevant to the aorta and similar tubular anatomical structures.
- **Task 3 - Implementation of State-of-Art Models:** Methodologies identified during the research phase were implemented. This process encompassed the development of baseline segmentation models and the extraction of geometric features from the aorta.
- **Task 4 - Writing the First Part of the Dissertation:** Following the literature review, the initial chapters of the dissertation were composed. This section covers the review of related work, a description of the selected dataset, and preliminary results derived from the proposed method.
- **Task 5 - Framework Development:** This stage involves the continued development of the computational framework, including a comparison with other state-of-the-art models and the refinement of feature extraction algorithms.
- **Task 6 - Writing the Second Part of the Dissertation:** Upon completion of the framework, the final phase involves documenting the full methodology and presenting a detailed analysis of the experimental results.

Bibliography

- [1] W. Oweini, J. Jolou, T. Sologashvili, N. Murith, C. Huber, and M. Cikirikcioglu, “Ascending aortic aneurysms: From pathophysiology to surgical repair,” *Journal of Clinical Medicine*, vol. 14, no. 19, 2025. [Online]. Available: <https://doi.org/10.3390/jcm14196993> 5
- [2] C. K. Cheng *et al.*, “Biophysical and biochemical roles of shear stress on endothelium: A revisit and new insights,” *Circulation Research*, vol. 136, no. 7, pp. 752–772, 2025. [Online]. Available: <https://doi.org/10.1161/CIRCRESAHA.124.325685> 5, 8
- [3] C. A. Beeche *et al.*, “Geometric aging of the thoracic aorta: Insights from 2 large cohorts,” *Hypertension*, vol. 82, no. 11, pp. 1866–1877, 2025. [Online]. Available: <https://doi.org/10.1161/HYPERTENSIONAHA.125.25259> 5
- [4] C. R. T. di Gioia *et al.*, “Thoracic aorta: Anatomy and pathology,” *Diagnostics*, vol. 13, no. 13, 2023. [Online]. Available: <https://doi.org/10.3390/diagnostics13132166> 6
- [5] A. Wanhainen *et al.*, “European society for vascular surgery (esvs) 2024 clinical practice guidelines on the management of abdominal aorto-iliac artery aneurysms,” *European Journal of Vascular and Endovascular Surgery*, vol. 67, no. 2, pp. 192–331, 2024. [Online]. Available: <https://doi.org/10.1016/j.ejvs.2023.11.002> 6, 8
- [6] D. Cai and S.-Y. Chen, “Elastin in the pathogenesis of abdominal aortic aneurysm,” *Cells*, vol. 14, no. 20, 2025. [Online]. Available: <https://doi.org/10.3390/cells14201597> 6
- [7] N. P. Ostberg *et al.*, “The genetics of thoracic aortic aneurysms and dissection: A clinical perspective,” *Biomolecules*, vol. 10, no. 2, 2020. [Online]. Available: <https://doi.org/10.3390/biom10020182> 6
- [8] E. M. Isselbacher *et al.*, “2022 acc/aha guideline for the diagnosis and management of aortic disease: A report of the american heart association/american college of cardiology joint committee on clinical practice guidelines,” *Circulation*, vol. 146, no. 24, pp. e334–e482, 2022. [Online]. Available: <https://doi.org/10.1161/CIR.0000000000001106> 7
- [9] D. J. Braet *et al.*, “Assessing differences in growth and shape between symptomatic and asymptomatic abdominal aortic aneurysms,” *Quantitative Imaging in Medicine and Surgery*, vol. 15, no. 7, 2025. [Online]. Available: <https://doi.org/10.21037/qims-2024-2985> 8
- [10] N. Kontopodis *et al.*, “Value of volume measurements in evaluating abdominal aortic aneurysms growth rate and need for surgical treatment,” *European Journal of Radiology*, vol. 83, no. 7, pp. 1051–1056, 2014. [Online]. Available: <https://doi.org/10.1016/j.ejrad.2014.03.018> 8

- [11] A. Vanmaele *et al.*, “The value of volume over maximum diameter for following abdominal aortic aneurysm growth and reducing surveillance visits in patients with a subthreshold aneurysm,” *Journal of Vascular Surgery*, vol. 82, no. 1, pp. 102–110.e3, 2025. [Online]. Available: <https://doi.org/10.1016/j.jvs.2025.03.395> 8
- [12] J. Malik *et al.*, “Wall shear stress alteration: a local risk factor of atherosclerosis,” *Current Atherosclerosis Reports*, vol. 24, pp. 143–151, 2022. [Online]. Available: <https://doi.org/10.1007/s11883-022-00993-0> 8
- [13] D. A. Vorp, “Biomechanics of abdominal aortic aneurysm,” *Journal of Biomechanics*, vol. 40, no. 9, pp. 1887–1902, 2007. [Online]. Available: <https://doi.org/10.1016/j.jbiomech.2006.09.003> 9
- [14] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” 2021. [Online]. Available: <https://doi.org/10.48550/arXiv.2010.11929> 11, 16, 17, 18
- [15] O. Ronneberger *et al.*, “U-net: Convolutional networks for biomedical image segmentation,” 2015. [Online]. Available: <https://doi.org/10.48550/arXiv.1505.04597> 14, 15, 21, 23
- [16] Z. Liu *et al.*, “Swin transformer: Hierarchical vision transformer using shifted windows,” 2021. [Online]. Available: <https://arxiv.org/abs/2103.14030> 17, 19
- [17] A. Hatamizadeh *et al.*, “Swin unetr: Swin transformers for semantic segmentation of brain tumors in mri images,” 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2201.01266> 18, 21, 23, 24
- [18] —, “Unetr: Transformers for 3d medical image segmentation,” 2021. [Online]. Available: <https://doi.org/10.48550/arXiv.2103.10504> 18, 21, 23, 24
- [19] A. Pepe *et al.*, “Detection, segmentation, simulation and visualization of aortic dissections: A review,” *Medical Image Analysis*, vol. 65, p. 101773, 2020. [Online]. Available: <https://doi.org/10.1016/j.media.2020.101773> 20, 22
- [20] Z. Sedghi Gamechi *et al.*, “Automated 3d segmentation and diameter measurement of the thoracic aorta on non-contrast enhanced ct,” *European Radiology*, vol. 29, no. 9, pp. 4613–4623, 2019. [Online]. Available: <https://doi.org/10.1007/s00330-018-5931-z> 21, 26
- [21] C. Mavridis *et al.*, “Automatic segmentation in 3d ct images: A comparative study of deep learning architectures for the automatic segmentation of the abdominal aorta,” *Electronics*, vol. 13, no. 24, 2024. [Online]. Available: <https://doi.org/10.3390/electronics13244919> 21, 24
- [22] A. Myronenko, “3d mri brain tumor segmentation using autoencoder regularization,” 2018. [Online]. Available: <https://doi.org/10.48550/arXiv.1810.11654> 21, 23

- [23] T. Kim *et al.*, “Computed tomography-based automated measurement of abdominal aortic aneurysm using semantic segmentation with active learning,” *Scientific Reports*, vol. 14, no. 1, p. 8924, 2024. [Online]. Available: <https://doi.org/10.1038/s41598-024-59735-8> 21, 24
- [24] N. Fink *et al.*, “Artificial intelligence provides accurate quantification of thoracic aortic enlargement and dissection in chest ct,” *Diagnostics*, vol. 14, no. 9, 2024. [Online]. Available: <https://doi.org/10.3390/diagnostics14090866> 21, 25
- [25] V. Kumari *et al.*, “Transformer and attention-based architectures for segmentation of coronary arterial walls in intravascular ultrasound: A narrative review,” *Diagnostics*, vol. 15, no. 7, 2025. [Online]. Available: <https://doi.org/10.3390/diagnostics15070848> 21
- [26] Y. Jin *et al.*, “Aortic vessel tree segmentation for cardiovascular diseases treatment: Status quo,” *ACM Comput. Surv.*, vol. 57, no. 9, Jun. 2025. [Online]. Available: <https://doi.org/10.1145/3728632> 22
- [27] A. Pepe *et al.*, *Segmentation of the Aorta: Towards the Automatic Segmentation, Modeling, and Meshing of the Aortic Vessel Tree from Multicenter Acquisition*, ser. Lecture Notes in Computer Science. Cham: Springer Nature Switzerland, 2024, vol. 14539. [Online]. Available: <https://doi.org/10.1007/978-3-031-53241-2> 22, 24
- [28] L. Radl *et al.*, “Avt: Multicenter aortic vessel tree cta dataset collection with ground truth segmentation masks,” *Data in Brief*, vol. 40, p. 107801, 2022. [Online]. Available: <https://doi.org/10.1016/j.dib.2022.107801> 22
- [29] A. Myronenko *et al.*, “Aorta segmentation from 3d ct in miccai seg.a. 2023 challenge,” in *Segmentation of the Aorta. Towards the Automatic Segmentation, Modeling, and Meshing of the Aortic Vessel Tree from Multicenter Acquisition*, A. Pepe *et al.*, Eds. Cham: Springer Nature Switzerland, 2024, pp. 13–18. [Online]. Available: https://doi.org/10.1007/978-3-031-53241-2_2 23, 24
- [30] M. J. Cardoso *et al.*, “Monai: An open-source framework for deep learning in healthcare,” 2022. [Online]. Available: <https://arxiv.org/abs/2211.02701> 23
- [31] Özgün Çiçek *et al.*, “3d u-net: Learning dense volumetric segmentation from sparse annotation,” 2016. [Online]. Available: <https://doi.org/10.48550/arXiv.1606.06650> 23
- [32] S. Liu *et al.*, “Segmentation-assisted vessel centerline extraction from cerebral ct angiography,” *Medical Physics*, vol. 52, no. 7, p. e17855, 2025. [Online]. Available: <https://doi.org/10.1002/mp.17855> 25
- [33] J. Vila Delgado, “Automatic feature extraction pipeline for abdominal aortic aneurysm characterization,” 2025. [Online]. Available: <https://hdl.handle.net/2445/221957> 25

- [34] R. Izzo, D. Steinman, S. Manini, and L. Antiga, "The vascular modeling toolkit: A python library for the analysis of tubular structures in medical images," *Journal of Open Source Software*, vol. 3, no. 25, p. 745, 2018. [Online]. Available: <https://doi.org/10.21105/joss.00745> 25, 26
- [35] R. Epifanov *et al.*, "The robust vessel segmentation and centerline extraction: One-stage deep learning approach," *Journal of Imaging*, vol. 11, no. 7, 2025. [Online]. Available: <https://doi.org/10.3390/jimaging11070209> 25
- [36] D. Bošnjak *et al.*, "A semi-automatic method for block-structured hexahedral meshing of aortic dissections," *International Journal for Numerical Methods in Biomedical Engineering*, vol. 40, no. 11, p. e3860, 2024. [Online]. Available: <https://doi.org/10.1002/cnm.3860> 26
- [37] P. Canals *et al.*, "A fully automatic method for vascular tortuosity feature extraction in the supra-aortic region: unraveling possibilities in stroke treatment planning," *Computerized Medical Imaging and Graphics*, vol. 104, p. 102170, 2023. [Online]. Available: <https://doi.org/10.1016/j.compmedimag.2022.102170> 27
- [38] J. L. Bruse *et al.*, "A statistical shape modelling framework to extract 3d shape biomarkers from medical imaging data: assessing arch morphology of repaired coarctation of the aorta," *BMC Medical Imaging*, vol. 16, no. 1, p. 40, 2016. [Online]. Available: <https://doi.org/10.1186/s12880-016-0142-z> 27
- [39] L. Geronzi *et al.*, "Computer-aided shape features extraction and regression models for predicting the ascending aortic aneurysm growth rate," *Computers in Biology and Medicine*, vol. 162, p. 107052, Aug. 2023. [Online]. Available: <http://dx.doi.org/10.1016/j.compbiomed.2023.107052> 27